

Processing State-of-the-Art Satellite Measurements with Machine Learning

Bachelor's Thesis of

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I declare that I have developed and written the enclosed thesis completely by myself. I have not used any other than the aids that I have mentioned. I have marked all parts of the thesis that I have included from referenced literature, either in their original wording or paraphrasing their contents. I have followed the by-laws to implement scientific integrity at KIT.

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Abstract

The analysis of satellite data for climate and weather prediction has become increasingly important in the context of global climate change. However, processing the vast amounts of data generated using traditional physical methods is computationally expensive. This thesis investigates the potential of machine learning techniques to partially replace full physics processing in the analysis of Infrared Atmospheric Sounding Interferometer (IASI)^{Gloss.} data, with a focus on improving computational efficiency while maintaining accuracy.

Various machine learning methods; Ridge Regression, Random Forest, Gaussian Process Regression (GPR) and Graph Convolutional Network (GCN), were applied to the IASI dataset. The performance of these models was evaluated in terms of accuracy and efficiency, i. e., energy consumption and run time, compared to the full physics processor. The results demonstrate significant efficiency gains, with processing times reduced by two to four orders of magnitude and energy consumption reduced by three to five orders of magnitude while maintaining reasonable accuracy in predicting key atmospheric variables.

The improved efficiency offers the potential for real-time or near real-time processing of atmospheric data, which could lead to more timely weather predictions and climate analysis. While full physics models remain essential for a comprehensive understanding of the atmosphere, the integration of machine learning techniques represents a significant step towards more efficient and responsive atmospheric data processing systems.

This thesis contributes to the ongoing efforts to manage and analyze the increasing volume of satellite data effectively, potentially enabling more comprehensive use of available data in weather forecasting and climate change research.

Zusammenfassung

Die Analyse von Satellitendaten für Klima- und Wettervorhersagen hat im Kontext des globalen Klimawandels zunehmend an Bedeutung gewonnen. Die Verarbeitung dieser enormen Datenmenge mit traditionellen physikalischen Methoden ist jedoch rechnerisch sehr aufwendig. Diese Arbeit untersucht das Potenzial von maschinellen Lernverfahren, um die vollständige physikalische Verarbeitung bei der Analyse von Daten des Infrared Atmospheric Sounding Interferometer (IASI)^{Gloss.} teilweise zu ersetzen, mit dem Ziel, die Recheneffizienz zu verbessern und zu gleich Genauigkeit zu erhalten.

Verschiedene Methoden des maschinellen Lernens; Ridge Regression, Random Forest, Gaussian Process Regression (GPR) und Graph Convolutional Network (GCN), wurden auf den IASI-Datensatz angewendet. Die Leistung dieser Modelle wurde hinsichtlich Genauigkeit und Effizienz im Vergleich zum vollständigen physikalischen Prozessor bewertet. Die Ergebnisse zeigen signifikante Effizienzsteigerungen, wobei die Verarbeitungszeiten um zwei bis vier Größenordnungen und der Energieverbrauch um drei bis fünf Größenordnungen reduziert wurden, während eine angemessene Genauigkeit bei der Vorhersage wichtiger atmosphärischer Variablen erhalten blieb.

Die verbesserte Effizienz bietet das Potenzial für eine Echtzeit- oder Nahezu-Echtzeit-Verarbeitung atmosphärischer Daten, was zu zeitnäheren Wettervorhersagen und Klimaanalysen führen könnte. Während vollständige physikalische Modelle für ein umfassendes Verständnis der Atmosphäre weiterhin unerlässlich sind, stellt die Integration von Techniken des maschinellen Lernens einen bedeutenden Schritt in Richtung effizienterer und reaktionsschnellerer atmosphärischer Datenverarbeitungssysteme dar.

Diese Arbeit trägt zu den laufenden Bemühungen bei, das zunehmende Volumen von Satellitendaten effektiv zu verwalten und zu analysieren, und ermöglicht potenziell eine umfassendere Nutzung der verfügbaren Daten in der Wettervorhersage und der Klimawandelforschung.

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1. Introduction

1.1. Motivation

In today's world, the study and understanding of our climate, its controlling factors, and their interactions have become more important than ever. This is particularly true in the context of climate change, a global issue with far-reaching implications for our planet and its inhabitants. The climate is a complex system, influenced by a multitude of factors that interact in intricate ways. One of the key aspects of this complex system is the atmospheric water cycle. The water cycle is a fundamental process that drives weather patterns and influences climate [22]. To understand this cycle and its effects on the climate, it is necessary to collect and analyze large amounts of data. Infrared Atmospheric Sounding Interferometer (IASI)^{Gloss.} is a space-based remote sensing instrument that collects data for this purpose. Currently, there are three IASI instruments in orbit, in total completing approximately 42 orbits every 24 hours, providing daily global coverage and resulting in 3.85 million individual IASI spectra per day [21]. IASI data has been collected since 2006. At KIT, the observations from 2014 to 2021 have been processed with full physics models, making optimal estimations for different atmospheric variables, resulting in approximately 40 TB of combined raw and processed data. The data from these satellites can be used to measure a large variety of atmospheric trace gas concentrations as well as humidity and temperature profiles. IASI data has many applications; it is being used to improve numerical weather prediction models [1], and the detection of climate change. However, processing the raw data with these physics models is computationally very expensive.

The goal of this thesis is to investigate to what extent the expensive processing of IASI data can be replaced using machine learning models. The hope is to improve computational efficiency with little or no loss in accuracy compared to the full physics model. This is essential as data collection is increasing at an unprecedented rate. Already today, we cannot fully process all the collected data using full physics models [12, p. 2]. These issues are made even worse by the need to reprocess older data, as models improve over time and allow for better products. Achieving significant improvements in computational efficiency could mean being able to process data in real time. This could lead to major improvements in all current and future applications of IASI data. Additionally, this could be extended to other products retrieved from IASI data, as well as other sensor data requiring expensive post-processing.

1.2. Related Work

There are two main categories of research related to this thesis: the full physics processing of the data and the use of machine learning methods to emulate physics models such as the radiative transfer^{Gloss.}.

1.2.1. Full Physics Processing

The full physics processing as described in “Optimal estimation of tropospheric H₂O and δ D with IASI/METOP” [22] is considered the gold standard for processing IASI data. However, it is very computationally expensive, primarily due to the need to calculate the radiative transfer models. In this thesis, we will consider the outputs of this full physics model as the truth and train models to emulate this processing.

1.2.2. Emulating Physics Models

Emulating expensive physics models, like the radiative transfer, has received significant attention in recent years to reduce computing runtime and energy consumption of models [2 | 23]. Typical approaches include neural networks and random forests, with many promising results reported.

In “Exploring Machine Learning to Correct Satellite-Derived Sea Surface Temperatures” [19], infrared remote sensing and ground “truth” from drifting buoy measurements were used to train different machine learning models to improve predictions of Sea Surface Temperatures (SST) from infrared remote sensing measurements. Random Forest showed the best performance in predicting the bias of satellite SST retrieval. However, this does not seem to have been done with other variables.

In “Machine learning applied to retrieval of temperature and concentration distributions from infrared emission measurements” [18] temperature and species concentrations for a gas mixture of CO₂, H₂O and CO were predicted from infrared spectral emission measurements using a neural network method. The method shows excellent capabilities while also being very efficient. However this was only done for synthesized data or data from combustion experiments in a controlled environment, and not for data across the entire atmosphere.

The European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) uses PWLR (Piece-Wise Linear Regression-cube) on IASI measurements to make “first guesses” of the atmospheric variables. These are then used in the full physics model for the optimal estimation [4]. These “first guesses” are provided in EUMETSAT Level 2 data and will also be used in this thesis.

In “A General Framework for Global Retrievals of Trace Gases From IASI” [6], IASI data is used to retrieve atmospheric trace gases (CH₃OH, HCOOH, and PAN) with a Neural Network approach.

1.3. Outline

Chapter 2 talks about the dataset that was used, the division into predictors and predictands, and the necessary preprocessing of the data.

Chapter 3 briefly introduces the different models used in this thesis. The models discussed include Ridge Regression, Random Forest Regression, Gaussian Process Regression, Graph Convolutional Networks.

Chapter 4 discusses the performance of these models when applied to the IASI dataset, the challenges encountered, and how they were solved. It also addresses the efficiency of each model and actions taken to utilize available resources more efficiently.

Chapter 5 presents the conclusion of this thesis.

2. About the Data

The dataset used in this thesis is the Multi-platform remote Sensing of Isotopologues for investigating the Cycle of Atmospheric water (MUSICA) Infrared Atmospheric Sounding Interferometer (IASI)^{Gloss.} dataset [20]. Currently, IASI observations between 2014 and 2021 have been generated with the MUSICA processing, resulting in a dataset size of about 40 TB. In this thesis, the dataset is divided into predictors and predictands, i. e. variables used for prediction and variables to be predicted. The dataset comprises the MUSICA retrieval data as well as data provided by the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) like the L1C spectral radiances [21]. The latter one comes from satellite observations from the IASI instruments aboard Meteorological Operational (Metop) satellites of the EUMETSAT. IASI is the core instrument of the three Metop satellites and is a Fourier-transform spectrometer that measures the infrared part of the electromagnetic spectrum at 841 frequencies. Its main purpose is the support of numerical weather predictions. It has a high spectral resolution of 0.5 cm^{-1} and a high signal-to-noise ratio as well as a horizontal resolution of 12 km [21].

Besides these EUMETSAT L1C spectral radiances, the dataset also contains EUMETSAT IASI level 2 information. These include surface emissivity at 20 frequencies as well as a priori values for surface temperature, atmospheric temperature, and atmospheric water vapor at 29 altitude levels. The a priori values are derived from different measurements made by the IASI, Advanced Microwave Sounding Unit - A (AMSU-A), and Microwave Humidity Sounder (MHS) instruments carried aboard the Metop satellites of EUMETSAT. They are generated with the Piece-Wise Linear Regression-cube algorithm and are used as a baseline for the full physics processor. They contain a lot of information about the seasonal cycle for each observation. Lastly, our target variables come from the MUSICA IASI retrieval, they are the product of the optimal estimation algorithm [21]. For this thesis, we will only look at the outputted surface temperature, atmospheric temperature, and atmospheric water vapor.

All atmospheric values are present as a profile with 29 bins, each bin representing a certain altitude interval, where bin 0 is the highest and bin 28 is the lowest in altitude, Figure 2.1 illustrates where in the atmosphere these levels are. The altitude levels can be inaccurate, due to uncertainty of the effective surface pixel ground altitude over complex terrain (e. g. mountains), therefore pressure levels were used as a better indicator of the actual altitude. Additionally, the dataset includes metadata such as the time, the geographical location, and the zenith angle, i. e. the satellite instrument viewing angle. Overall, this adds up to 953 predictor and 59 predictand variables for each observation.

Table 2.1.: List of all Predictors and Predictands. Predictors are variables that will be used to predict the values of our target values (Predictands). Predictands are the target values that we want to predict using the predictors.

Predictors	
Time	UTC in seconds since 2000-01-01 00:00:00
Longitude	The geographical longitude in degree east
Latitude	The geographical latitude in degree north
Zenith Angle	The satellite instrument viewing angle, i.e. the angle between the satellite line of sight and the zenith ^{Gloss.}
Surface Emissivity	The surface emissivity at 20 different frequencies
Atmospheric Pressure	The atmospheric pressure at 29 different altitude levels
Spectra	The spectra measured by IASI at 841 frequencies between 1190 and 1400cm ⁻¹
Surface Temperature a priori	The a priori surface skin temperature as provided by EUMETSAT L2
Atmospheric Temperature a priori	The a priori atmospheric temperature as provided by EUMETSAT L2 at 29 different altitude levels
Water Vapor a priori	The a priori water vapor as provided by EUMETSAT L2 at 29 different altitude levels
Predictands	
Surface Temperature	The retrieved surface skin temperature in Kelvin
Atmospheric Temperature	The retrieved atmospheric temperature in Kelvin
Atmospheric Water Vapor	The retrieved atmospheric water vapor in ppm

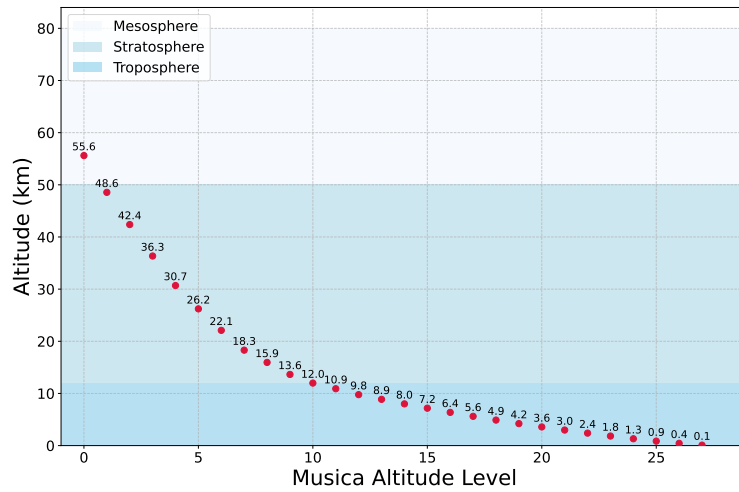


Figure 2.1.: Vertical distribution of atmospheric observation levels in January 2021. The plot shows the average altitude of each level, with the highest observations occurring at approximately 55 km above mean sea level, within the lower mesosphere. Levels one to ten are located in the stratosphere, while the remaining lower levels are situated in the troposphere. Atmospheric layers are color-coded for clarity.

2.1. Exemplary Observations

This section shows visualizations for most of the variables. For the variables with altitude profile, three observations over Germany in the year 2021 were chosen. Figures showing a complete orbit were made using one orbit from June 2021.

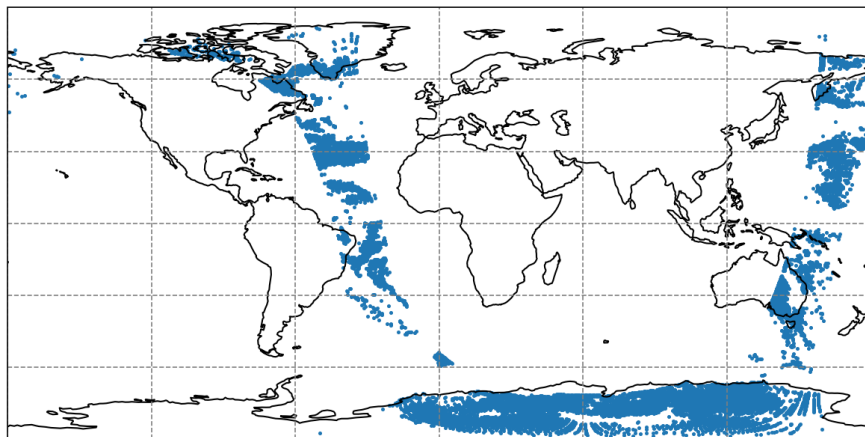


Figure 2.2.: Coordinates for all observations shown on a map for one exemplary orbit. As all satellites fly in a polar orbit^{Gloss.}, the orbits look alike but are shifted in longitude, resulting in global coverage.

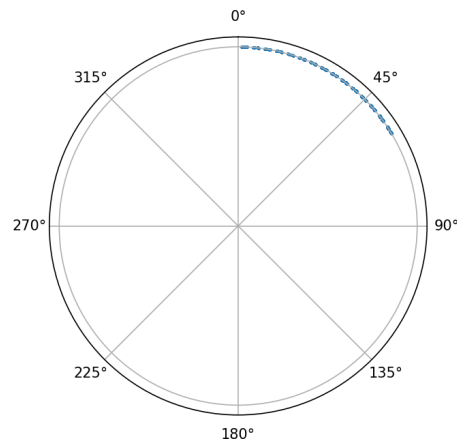


Figure 2.3.: Zenith angles for all observations during one orbit i. e. the satellite instrument viewing angle. The values range from 0° to 60° in this dataset.

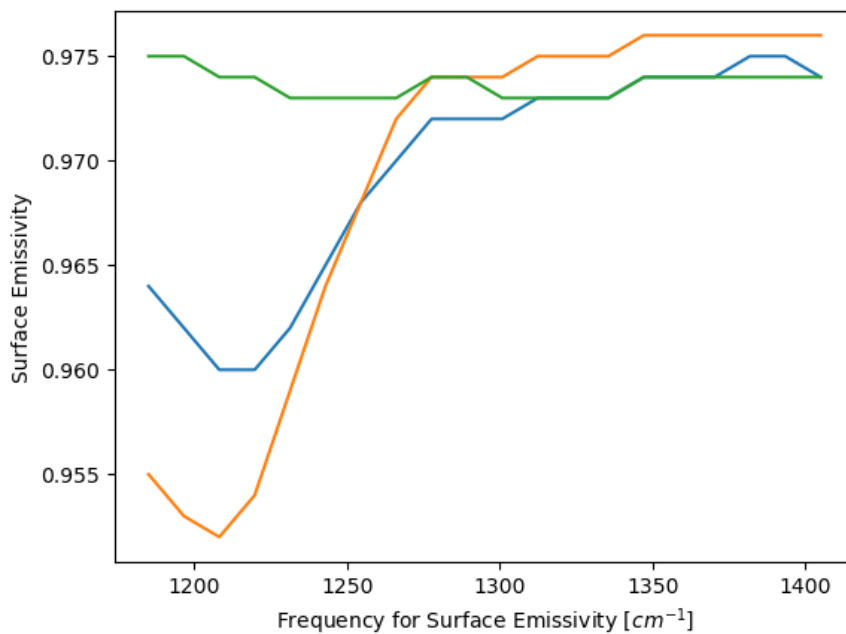


Figure 2.4.: Surface emissivity at each frequency for three exemplary observations. The biggest differences can usually be observed in the lower frequencies. Lower surface emissivity, especially in the lower frequencies, is typical for dry desert areas (like the Sahara, Australia), and thus usually correlates with higher surface temperatures.

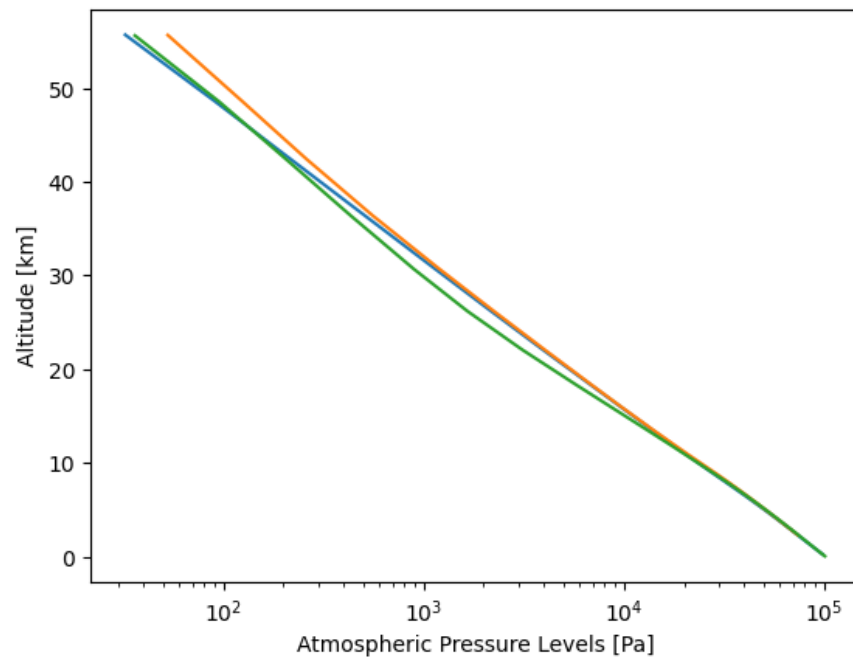


Figure 2.5.: Atmospheric pressure profiles from three observations plotted against their altitudes. The altitude profiles consist of 29 bins, with pressure serving as a more precise indicator of altitude. It can be seen that altitude and atmospheric pressure are highly correlated and there should not be a high variance between observations. Note: Because of this relationship high y values will correspond to low altitudes in the following plots. The x scale is logarithmic.

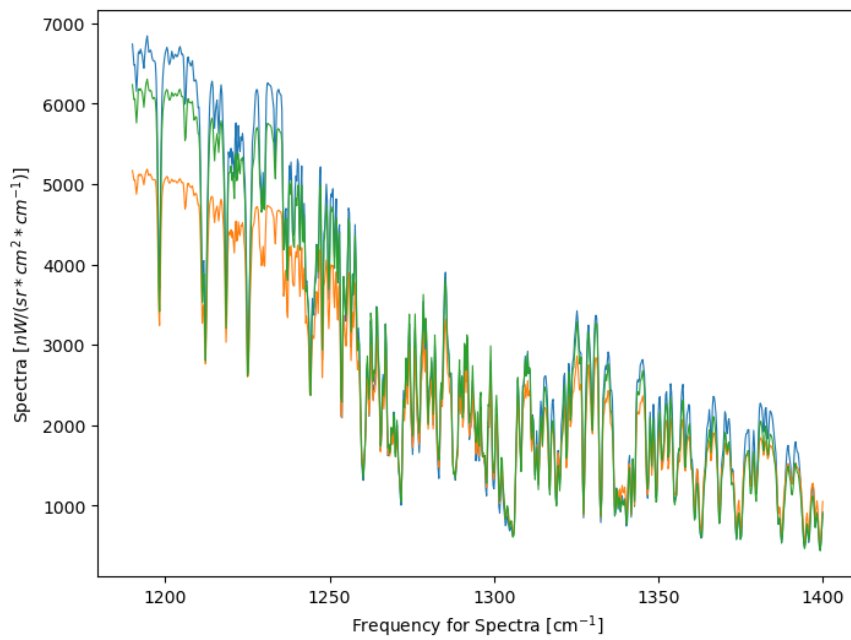


Figure 2.6.: IASI spectra by MUSICA IASI processor for three exemplary observations. The highest variance can usually be observed in the lower frequencies. Below about 1200cm^{-1} the radiation is only weakly affected by the atmosphere, i. e. the intensity is strongly correlated to the surface temperature. High surface temperatures are typical for low latitudes, where the atmospheric water vapor content is also typically high.

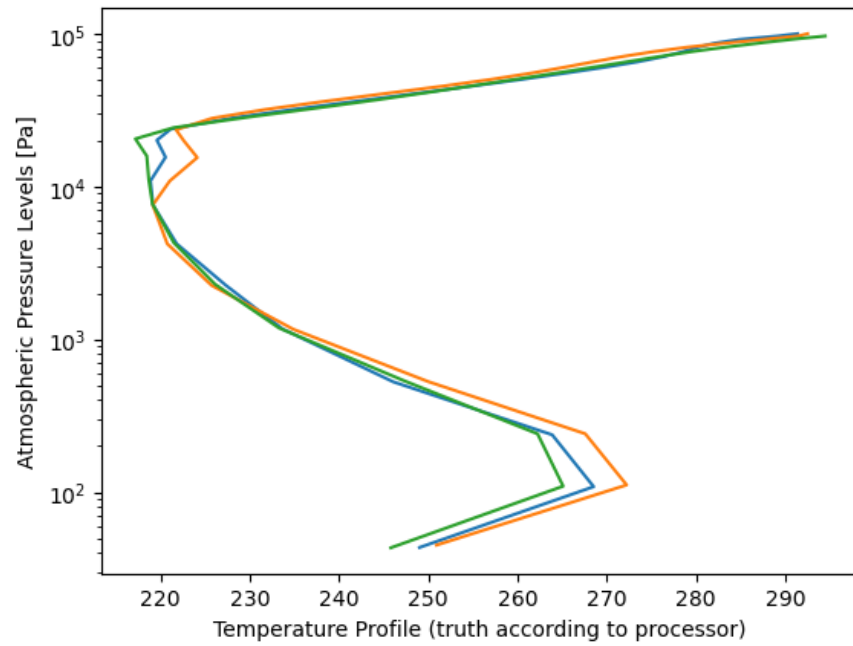


Figure 2.7.: A priori temperature profiles in Kelvin with EUMETSAT L2 as source at different altitude levels for three exemplary observations. Atmospheric pressure is used as an indicator of the actual altitude. It is on a log scale as the relation to the altitude is exponential due to exponential decay of pressure with altitude, i. e. high pressure means low altitude, low-pressure high altitude (see 2.5).

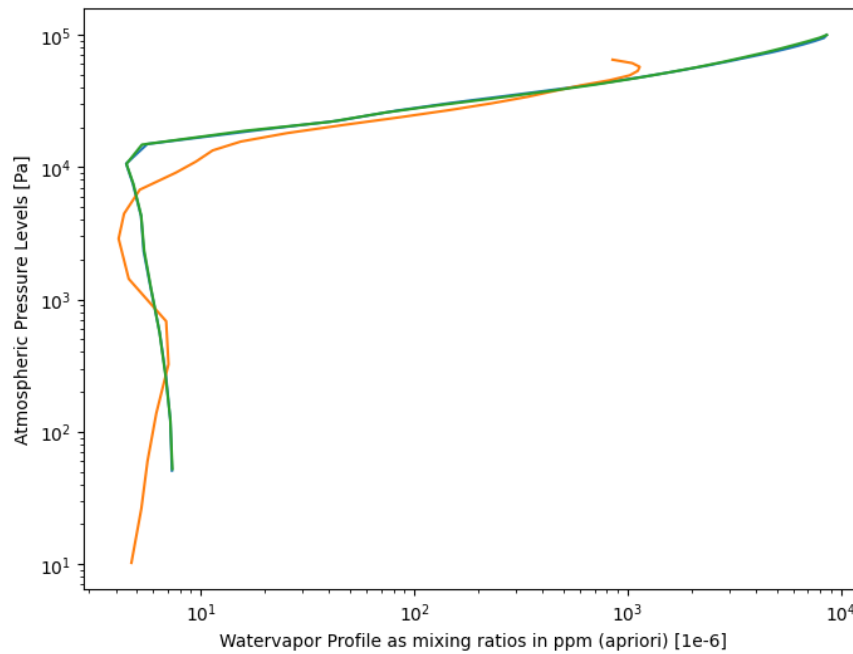


Figure 2.8.: A priori water vapor profiles as mixing ratios in parts per million (ppm) for three exemplary observations. As can be seen here, water vapor values are largest for the highest pressure values, i. e. close to the ground.

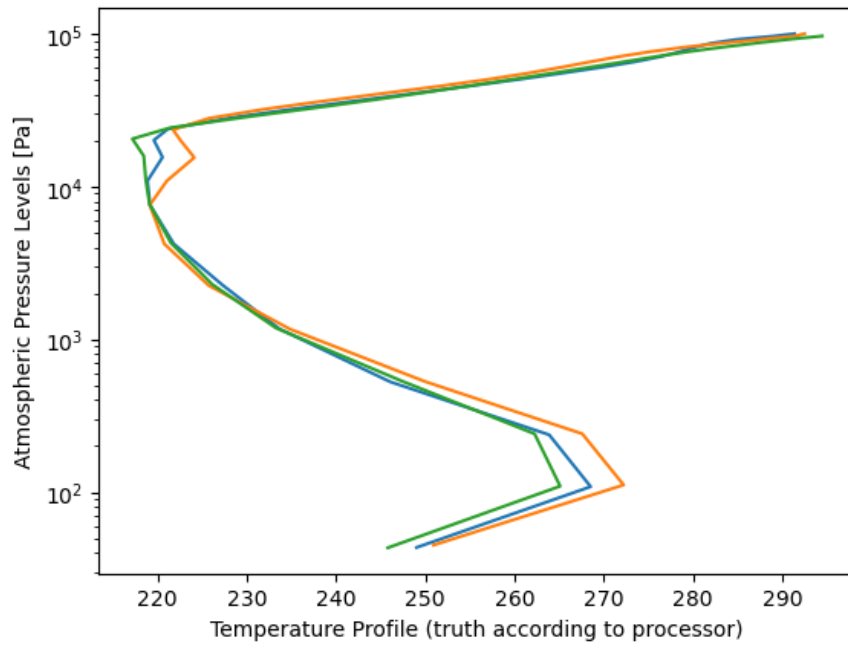


Figure 2.9.: Retrieved atmospheric temperature profiles in Kelvin for three exemplary observations. The retrieved values are expected to be in the same order of magnitude as the a priori values.

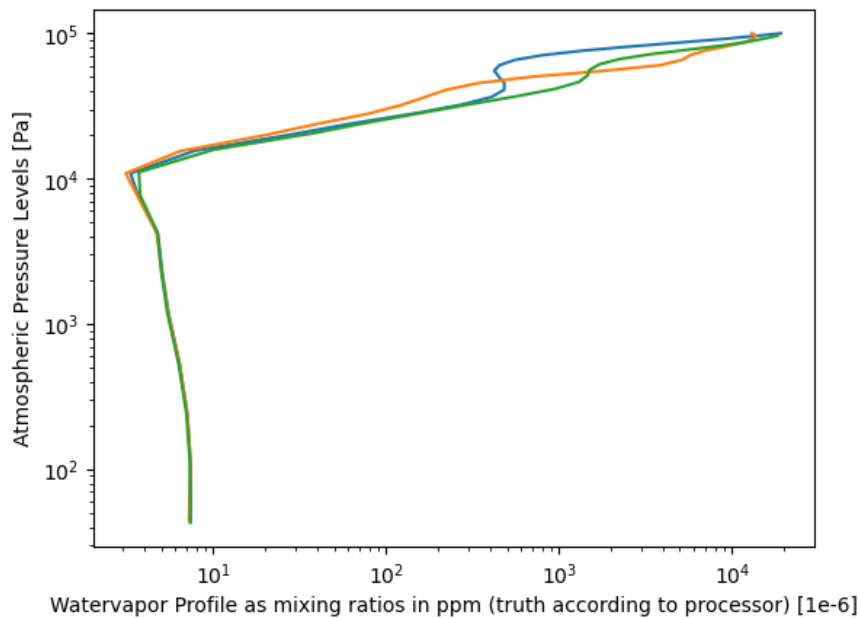


Figure 2.10.: Retrieved profiles of water vapor profiles as mixing ratios in ppm for three exemplary observations. The retrieved values are expected to be in the same order of magnitude as the a priori values.

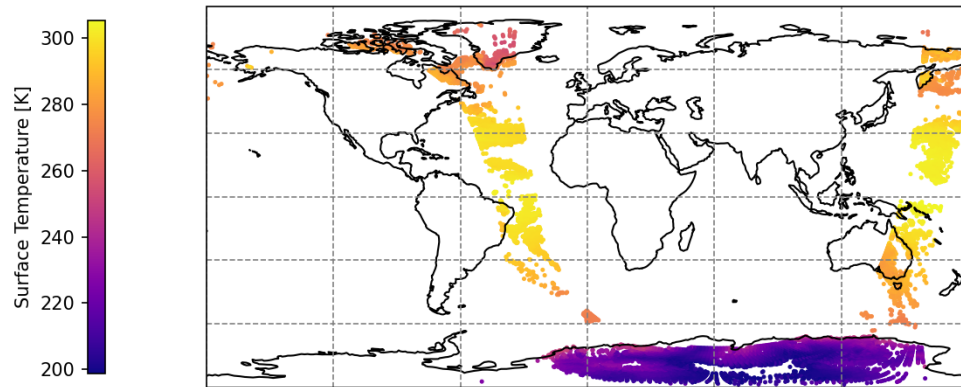


Figure 2.11.: Map with the retrieved surface skin temperature in Kelvin for all observations of one orbit.

2.2. Data Selection and Preprocessing

In the MUSICA dataset, the spectra are only published as spectra simulations and spectra residuals. We wanted to learn from the raw spectra so we calculated these as the difference between simulated spectra and their residuals. As the ground altitude varies, some of the 29 atmospheric levels don't contain any data. To avoid NaN values in the training data, a forward-fill was used i. e. the latest valid measurement was used for all following levels on all affected rows. Alternatively, this could have been solved by setting them to a default value e. g. the mean of each level. For this thesis, we only work with H₂O and disregard HDO. Both vary largely in parallel and the variabilities of the HDO/H₂O ratios are of a secondary order.

The most up-to-date processed data at the time of this work was for the years 2019 (123 762 657 observations), 2020 (306 818 090 observations), and half of the year 2021 (135 223 109 observations). The dataset for the year 2020 was the biggest, which is why it was chosen as the training dataset, 2019 being the second dataset to cover the entire year was used as the validation dataset, and 2021 as the testing dataset.

As the dataset is very big, the quality of observations differs a lot, and not all data can be used during training, mostly due to memory limitations, we have filtered the data according to a few features. Firstly we only used observations that had a fit quality flag of 3, the fit-quality flag is outputted by the full-physics processor, and a value of three indicates the best quality of data. We also removed all observations that had a relative humidity higher than 200%, as values higher than 100% are not physically possible and indicate large uncertainties in the respective full physics result. The relative humidity was calculated with:

$$\begin{aligned} \text{over liquid: } RH &= wv \cdot \left(10^2 \cdot 10^6 \cdot 6.11 \cdot \frac{1}{\text{pressure}} \cdot \exp \left(17.62 \cdot \frac{at - 273.15}{243.12 + (at - 273.15)} \right) \right)^{-1} \\ \text{over ice: } RH &= wv \cdot \left(10^2 \cdot 10^6 \cdot 6.11 \cdot \frac{1}{\text{pressure}} \cdot \exp \left(22.46 \cdot \frac{at - 273.15}{272.62 + (at - 273.15)} \right) \right)^{-1} \end{aligned}$$

This high-quality data filtering left about 78% of the original data. Since there were still too many data points we used steps to take only every 10th, 100th, 1000th, or 10 000th data point depending on the space and time complexity of the model we were currently working on.

3. Preliminaries

This chapter introduces the machine learning methods employed in subsequent analyses. The techniques discussed are Ridge Regression, Random Forest Regression, Gaussian Process Regression (GPR), and Graph Convolutional Networks (GCNs). These methods are selected for their ability to model diverse data relationships and structures. Ridge Regression addresses multicollinearity in linear models. Random Forest Regression handles non-linear relationships through ensemble learning. GPR offers a probabilistic approach to regression. GCNs apply deep learning to graph-structured data. The basic principles of each method are presented to establish a foundation for their application in later chapters.

3.1. Ridge Regression

Ridge Regression is a method used to handle ill-conditioned input matrices in linear regression models. The Ordinary Least Squares (OLS) method can be expressed as:

$$\underset{\beta}{\text{minimize}} \|y - X\beta\|_2^2$$

where X is the input matrix, y is the output vector and β are the parameters. This problem can be ill-posed when the predictor variables are highly correlated and can cause unstable and unreliable parameter estimates. Ridge regression addresses this by adding a penalty term to the standard model for multiple linear regression, which shrinks the parameters and reduces their variance. The resulting optimization problem can be expressed as

$$\underset{\beta}{\text{minimize}} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2, \quad \lambda > 0$$

where λ is a hyperparameter^{Gloss.} influencing the strength of the regularisation. Introducing λ comes at the cost of a bias towards zero and a smaller variance than OLS, i.e. without the penalty term [9]. The parameters can be calculated with $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$. The matrix inversion is the most expensive operation and can be solved using singular value decomposition^{Gloss.}. Therefore calculating optimal parameters can be done in $O(mn^2)$, where n is the number of observations and m is the number of features.

3.2. Random Forest Regression

Random Forest Regression is an ensemble learning method that enhances tree bagging. In Tree Bagging multiple decision trees are constructed during training. The average of the predictions of these trees is the output. The inner nodes of a decision tree are decision nodes,

where one of the children is selected based on a feature and a threshold. The leaves of the tree are terminal nodes and in the case of regression contain a continuous value. In regression trees, this value is typically the mean of the target variable for the training samples that fall into that leaf.

Random Forest enhances simple Tree Bagging by using bootstrapping to build the trees. Bootstrapping is a resampling technique where random samples are drawn (with replacements) from the original dataset to create multiple subsets. Each of these subsets is then used to train an individual decision tree.

The Random Forest algorithm also employs Classification and Regression Trees (CART) as its base learners. CART is a decision tree algorithm that can handle both classification and regression tasks. In regression, CART recursively partitions the feature space to minimize the mean squared error at each split. This process continues until a stopping criterion is met, such as reaching a maximum depth or a minimum number of samples in a leaf [8].

Training a Random Forest is done in $O(kn \log(n)m)$ where k is the number of trees, n is the number of observations, and m is the number of features. As the trees are independent of one another, they can be trained in parallel. Hyperparameters for Random Forest Regression include the number of decision trees in the forest, the maximum possible depth of each tree, the maximum number of features the model will consider when determining a split, and the minimum number of samples required to make a split.

3.3. Gaussian Process Regression

Gaussian Process Regression (GPR) is a non-parametric, probabilistic machine learning method. By leveraging the properties of Gaussian processes and Bayesian inference, it can model complex relationships while quantifying uncertainty in predictions.

In GPR, we assume that the data can be sampled from a Gaussian process, which is a generalization of the multivariate Gaussian distribution. If a random variable $x \in \mathbb{R}^n$ follows a multivariate Gaussian distribution with mean μ and covariance K , we write $x \sim \mathcal{N}(\mu, K)$, that is if

$$p(x, \mu, K) = \frac{1}{(2\pi)^{n/2} |K|^{1/2}} \exp \left(-\frac{(x - \mu)^T K^{-1} (x - \mu)}{2} \right).$$

Gaussian Processes generalize this to an infinite number of dimensions which can be viewed as a distribution over functions. This is achieved by using a mean function $m(\cdot)$ and a covariance kernel $k(\cdot, \cdot)$, for a finite subset we can write the mean as [7]

$$\mu = \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix} \quad \text{and the covariance as} \quad K = \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix}$$

In GPR we usually also assume a zero mean function. Therefore following applies to our target $y \sim \mathcal{N}(0, K)$. The conditional probability $p(y_*|y)$, can be used to make new predictions y_* , this conditional probability also follows a Gaussian distribution, with the closed form being

$$y_*|y \sim \mathcal{N}\left(K_*K^{-1}y, K_{**} - K_*K^{-1}K_*^T\right)$$

where $K_* = \begin{bmatrix} k(x_*, x_1) & \cdots & k(x_*, x_n) \end{bmatrix}$ and $K_{**} = k(x_*, x_*)$. The best prediction is the mean of this distribution $K_*K^{-1}y$ and the uncertainty is represented by its variance $K_{**} - K_*K^{-1}K_*^T$ [3].

Hyperparameters of GPR include the choice of kernel function, kernel parameters (such as length scales), and the noise variance. Kernel functions measure the similarity between two data points in the input space. In GPR, they define the covariance between function values at different input points. The choice of kernel determines the properties of the functions that can be modeled by the Gaussian Process. Kernels can be combined through addition or multiplication to create more complex covariance structures. A common kernel function is the Radial Basis Function (RBF)

$$k(x_1, x_2) = \exp\left(-\frac{\|x_1 - x_2\|_2^2}{2l^2}\right)$$

where l is the lengthscale. Gaussian Processes using this kernel are very smooth. The Matérn kernel is a generalisation of RBF with the additional parameter $\nu > 0$ (often $\frac{3}{2}$ or $\frac{5}{2}$) which controls the smoothness. It is given by

$$k(x_1, x_2) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}\|x_1 - x_2\|}{l}\right)^\nu K_\nu\left(\frac{\sqrt{2\nu}\|x_1 - x_2\|}{l}\right)$$

where Γ is the gamma function and K_ν is a modified Bessel function [17]. A simple white noise kernel

$$k(x_1, x_2) = \text{noise_le} \text{ if } x_1 == x_2 \text{ else } 0$$

can be added to other Kernels and explains the noise in the data.

GPRs time complexity lies in $O(n^3)$ and the space complexity lies in $O(n^2)$, by using sparse Gaussian processes the time complexity can be reduced to $O(nm^2)$, with m being the number of inducing points [16 | 17]. Inducing points act as a low-rank approximation of the full covariance matrix. They can be thought of as a representative subset of the input space that captures the essential structure of the full dataset.

3.4. Graph Convolutional Networks

In the field of machine learning on graph-structured data, Graph Convolutional Network (GCN) have gained significant attention. GCNs are a type of Graph Neural Network (GNN) that can effectively process both node features and graph topology to make predictions.

Formally, a graph G is defined as a tuple $G = (V, E)$, where V is the set of vertices (or nodes) and E is the set of edges connecting these vertices. This mathematical representation allows for the modeling of various real-world systems and relationships, from social networks to molecular structures.

GCNs have found applications in diverse areas such as physical simulations, social network analysis, and recommendation systems. Their ability to capture complex relationships within graph-structured data has made them particularly useful in scenarios where traditional machine-learning approaches may fall short.

A multi-layer GCN can be formulated using the layer-wise propagation rule [11]:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right)$$

where $H^{(l)}$ represents the node features at the l -th layer, $\tilde{A}$ is the adjacency matrix with added self-connections, $\tilde{D}$ is the degree matrix of $\tilde{A}$ used to normalize $\tilde{A}$ to avoid changing feature scale with each multiplication, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. $W^{(l)}$ is the layer-specific trainable weight matrix, and $\sigma(\cdot)$ denotes an activation function e. g. $\text{ReLU}^{\text{Gloss}}$. In this formulation, the graph structure is encoded in the $\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}$ term, which can be precomputed for efficiency. The final layer in this architecture is typically a linear output layer. Its forward propagation is given by

$$Y = H^{(n)} W + b$$

where Y is the output, $H^{(n)}$ is the last hidden representation, b is the bias vector and W are the learnable weights.

The power of GCNs lies in their ability to efficiently learn representations that capture local graph structure. Each layer in a GCN aggregates information from a node's immediate neighbors. By stacking multiple layers, the network can incorporate information from a wider neighborhood. However, they subsequently have their limitations in capturing long-range dependencies or global graph structure.

4. Evaluation

In this chapter, the performance of four regression models is evaluated: Ridge Regression, Random Forest Regression, Gaussian Process Regression, and Graph Convolutional Networks. These models were applied to predict surface temperature, atmospheric temperature, and water vapor levels based on Infrared Atmospheric Sounding Interferometer (IASI)^{Gloss} measurements to approximate the output of full-physics models.

The evaluation methodology employs R^2 scores, also known as the coefficient of determination, to quantify the models' predictive accuracy. R^2 values range from $-\infty$ to 1, with 1 indicating perfect prediction and 0 indicating that the model performs no better than a prediction of the mean. This metric provides a measure of how well the regression predictions approximate the real data points. Additionally, scatter plots comparing the true values (i. e. the full-physics model output) to the predicted values for each variable are utilized to visually assess model performance.

To achieve optimal performance, input normalization was necessary for all models. This preprocessing step ensures that all features contribute equally to the model and accelerates convergence during training. An important constraint was applied to the models' outputs. All predicted values smaller than 0 were set to 0, as negative values for the variables under consideration are physically impossible, e. g. temperature below 0 Kelvin.

The prediction tasks varied in difficulty across different atmospheric variables. Temperature was expected to have an approximately linear relationship with the input variables, making it easy to predict. In contrast, water vapor was anticipated to have an exponential relationship with the inputs, presenting a more challenging prediction task. Consequently, later iterations of the modeling process focused more intensively on improving water vapor predictions.

In this chapter, we aim to assess the strengths and limitations of each regression method in approximating the full-physics model. This comparative analysis will provide insights into which approaches are most suitable for different aspects of atmospheric prediction, potentially guiding future work in this field.

4.1. Ridge Regression

For the Ridge Regression model, a gridsearch approach was employed to determine the optimal regularization parameter for each output. A key improvement was achieved by learning only the residuals to the a priori values, which effectively removed biases in the atmospheric temperature predictions. Additional preprocessing techniques were experimented with, including normalization of the outputs and a logarithmic transformation on all values related to water vapor. Applying a logarithmic transformation to the water vapor and its

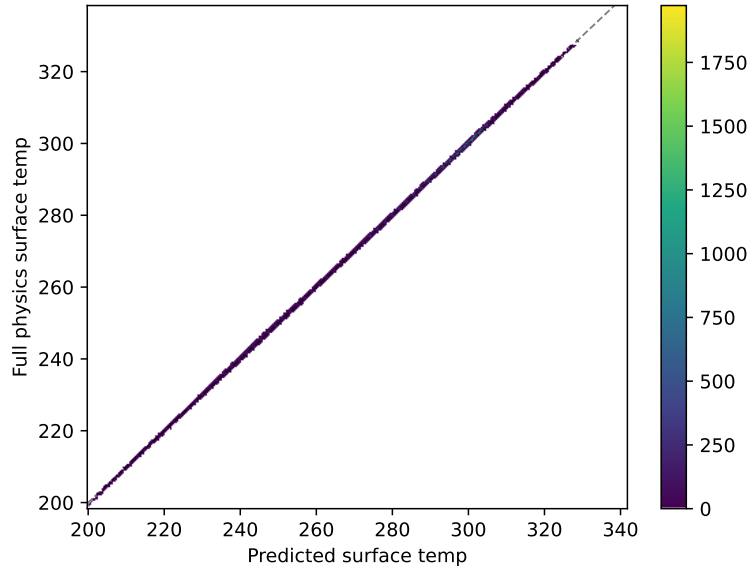


Figure 4.1.: Predicted over Actual normalized Surface Temperature using Ridge Regression. The model achieves an R^2 score of 0.9998, demonstrating excellent predictive performance for surface temperature. The color scale shows points per pixel.

a priori values did improve performance on the water vapor prediction for atmospheric levels seven to seventeen, but decreased the predictive performance in the other levels as well as the performance of temperature predictions.

The final results achieved were better than expected with R^2 values of 0.9998 for surface temperature, 0.9997 for atmospheric temperature, and 0.86897 (0.8830 with log transformation) for water vapor. Figure 4.2 shows the scatter plot comparing the true values to the predicted values. It can be seen that the model is performing best for the atmospheric temperature at the highest levels (i. e. upper row of plots), which is where the relationships are expected to be most linear. The upper levels are also where the model performs worst for water vapor predictions, which is where the relationship is expected to be exponential. By far the worst predictions are made in level eight. This level lies in the lower troposphere, right at the start of the ozone layer, which might be why making predictions is so difficult.

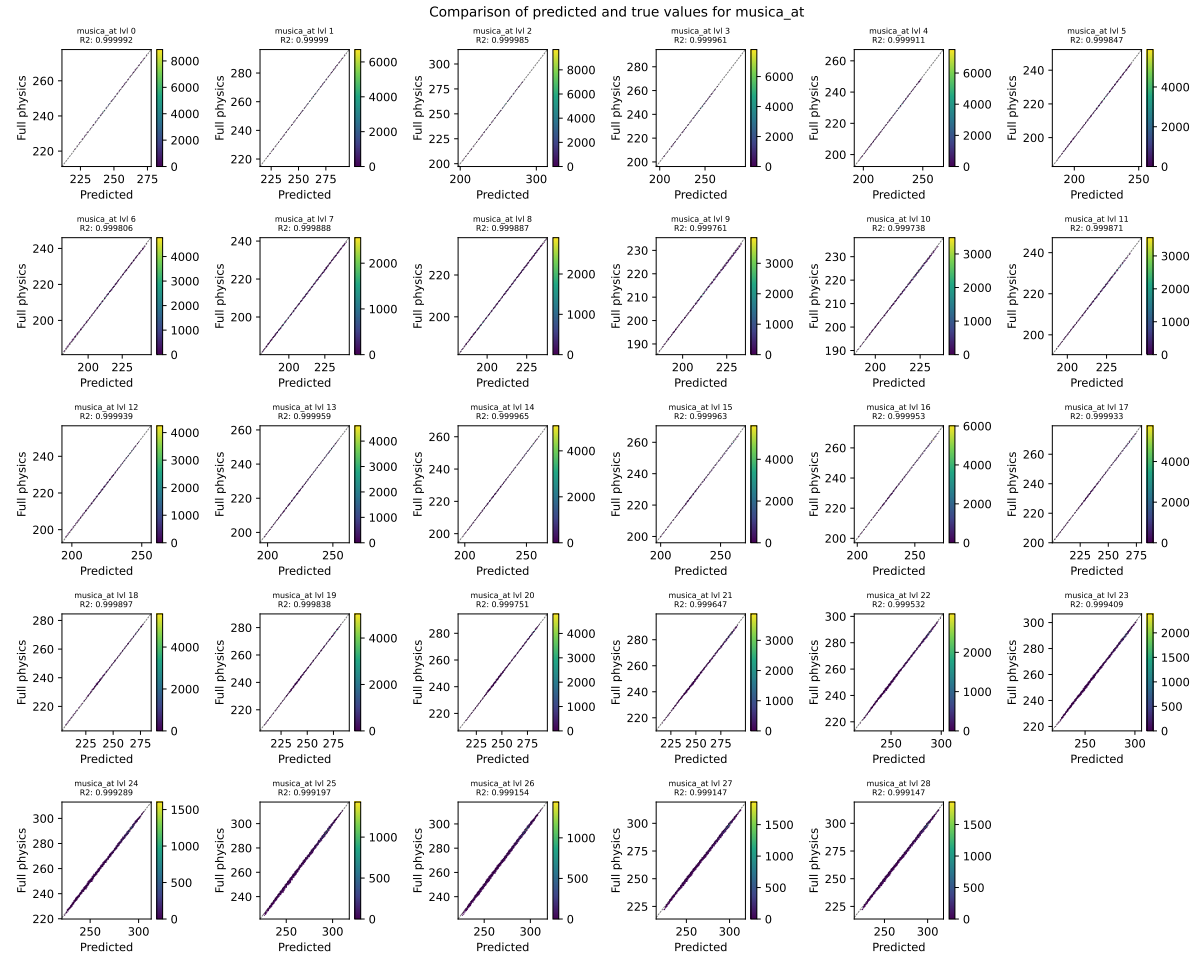


Figure 4.2.: Predicted over Actual normalized Atmospheric Temperature across different atmospheric levels using Ridge Regression. The model attains an average R^2 score of 0.9997, with the best performance observed in the highest atmospheric levels where relationships are expected to be most linear. The color scale shows points per pixel.

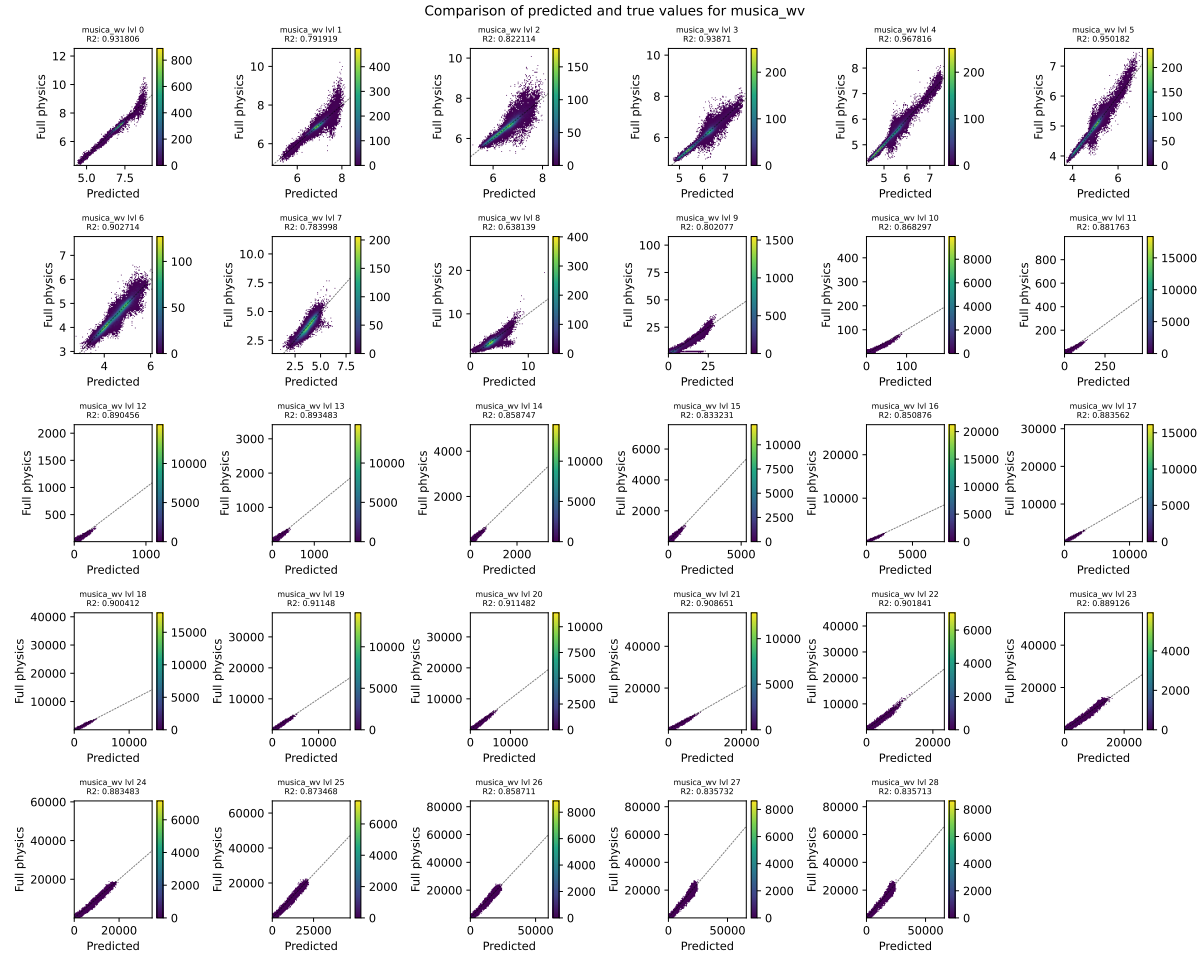


Figure 4.3.: Predicted over Actual normalized Water Vapor Levels across atmospheric layers using Ridge Regression. The model achieves an average R^2 score of 0.86897. Performance is notably worse than for atmospheric temperature due to expected exponential relationships. This is particularly true in the stratosphere where water vapor is a lot more rare than in the troposphere. The color scale shows points per pixel.

4.2. Random Forest Regression

The Random Forest Regression model demonstrated significant performance improvements over the previously presented Ridge Regression model, particularly when employing separate forests for each output variable. This approach led to notable enhancements in prediction accuracy, especially for water vapor. Using Optuna [14], a hyperparameter^{Gloss.} optimization framework, several key insights into model behavior and performance were found.

One of the most important discoveries was that trees dedicated to water vapor prediction benefited substantially from increased sizes, while those focused on temperature prediction did not show similar gains. Increasing the maximum depth of the trees much higher than 15 showed diminishing returns, although increasing depth did not hurt performance. The minimum number of samples required to split a node showed no significant impact on performance and should probably be kept low to avoid unnecessary constraints.

The use of individual forests for each output variable, while highly effective, did lead to a considerable increase in model size. This approach resulted in the largest model in this chapter's comparison. Despite the increased computational demands, the performance gains justified the trade-off.

The best Random Forest Regression model achieved better results for surface temperature and water vapor profiles while performing slightly worse on atmospheric temperature predictions. For surface temperature, we obtained an R^2 value of 0.999 932, this is visualized in Figure 4.4, demonstrating near-perfect predictive performance. Atmospheric temperature prediction was similarly strong, with an R^2 value of 0.999 146. The slight decrease in performance compared to Ridge Regression, is probably due to Random Forest's incapability of perfectly representing linear relationships. In Figure 4.5, we observe that in the lower atmospheric levels, the model appears to struggle with predicting extremely low temperatures. This may indicate limitations in the model's ability to extrapolate beyond the range of its training data. Water vapor predictions, while still challenging, also showed better results with an R^2 value of 0.925 927. This represents a substantial improvement over the simpler model Ridge Regression but still presents some challenges. As illustrated in Figure 4.6, the model's performance varies across atmospheric layers, with levels seven and eight proving particularly difficult to predict accurately. This pattern of difficulty aligns with observations from other models, suggesting an intrinsic challenge in modeling water vapor at these specific atmospheric levels.

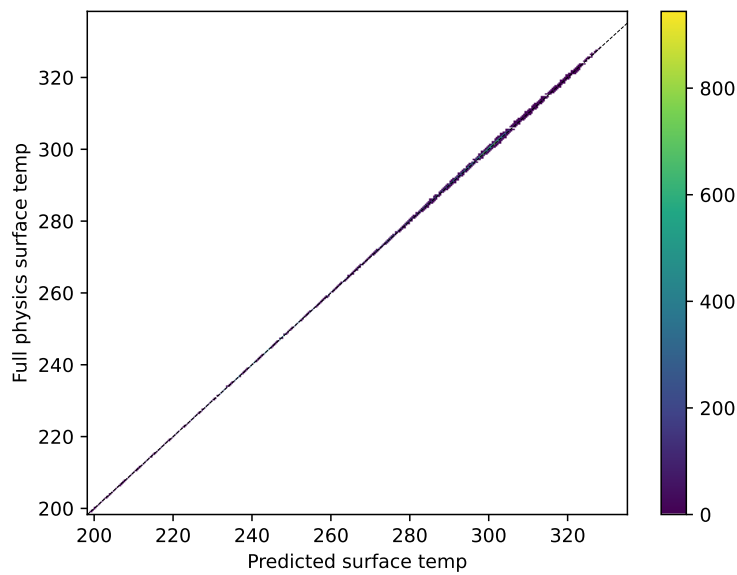


Figure 4.4.: Predicted over Actual Surface Temperature using Random Forest Regression. The model achieves an R^2 score of 0.999 932, demonstrating excellent predictive performance for surface temperature. The color scale shows points per pixel.

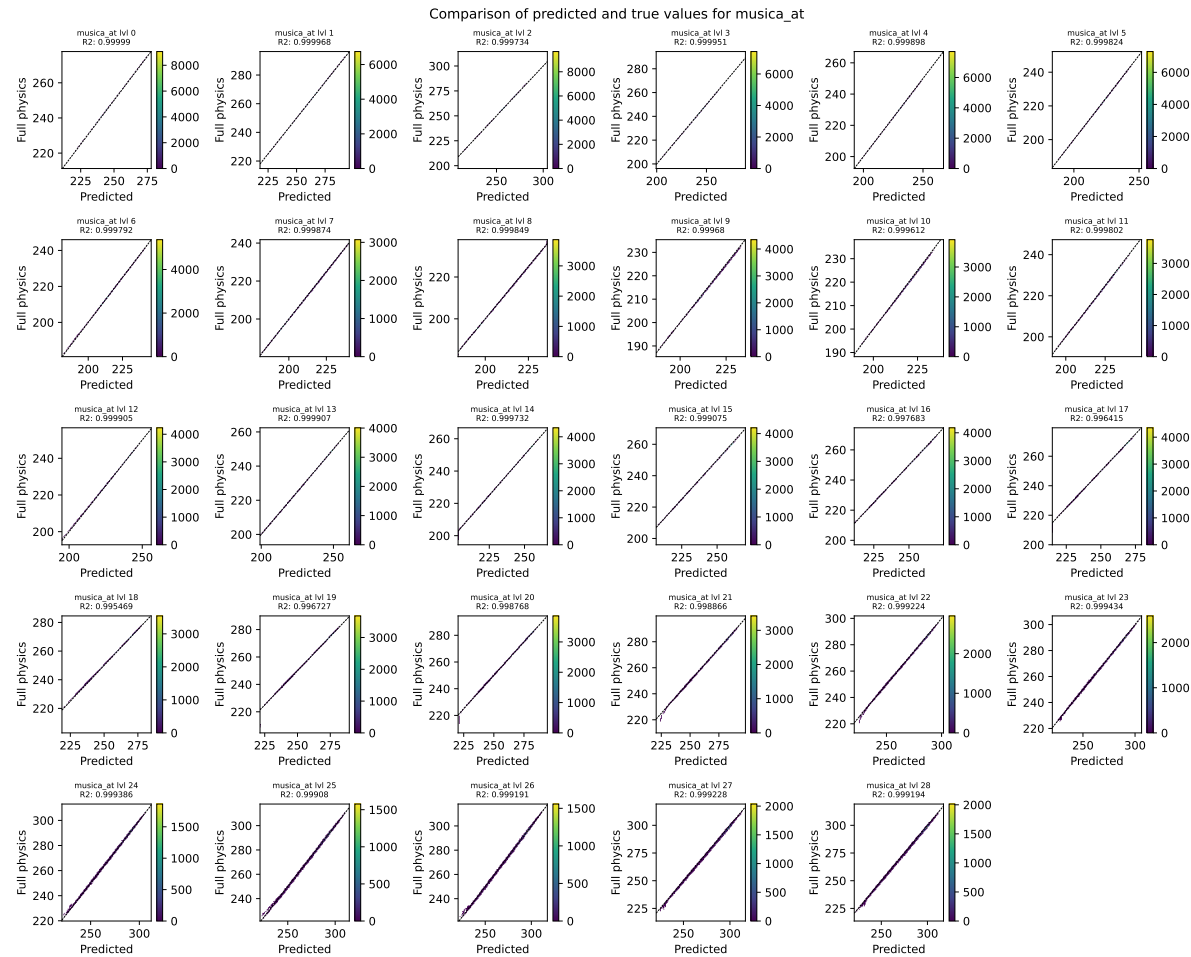


Figure 4.5.: Predicted over Actual Atmospheric Temperature across different atmospheric levels using Random Forest Regression. The average R^2 score of 0.999 146 tells us that the model overall can predict the atmospheric temperature very well. In the lower atmospheric levels, predictions for extremely low temperatures seem to be challenging, this might indicate poor extrapolation skills. The color scale shows points per pixel.

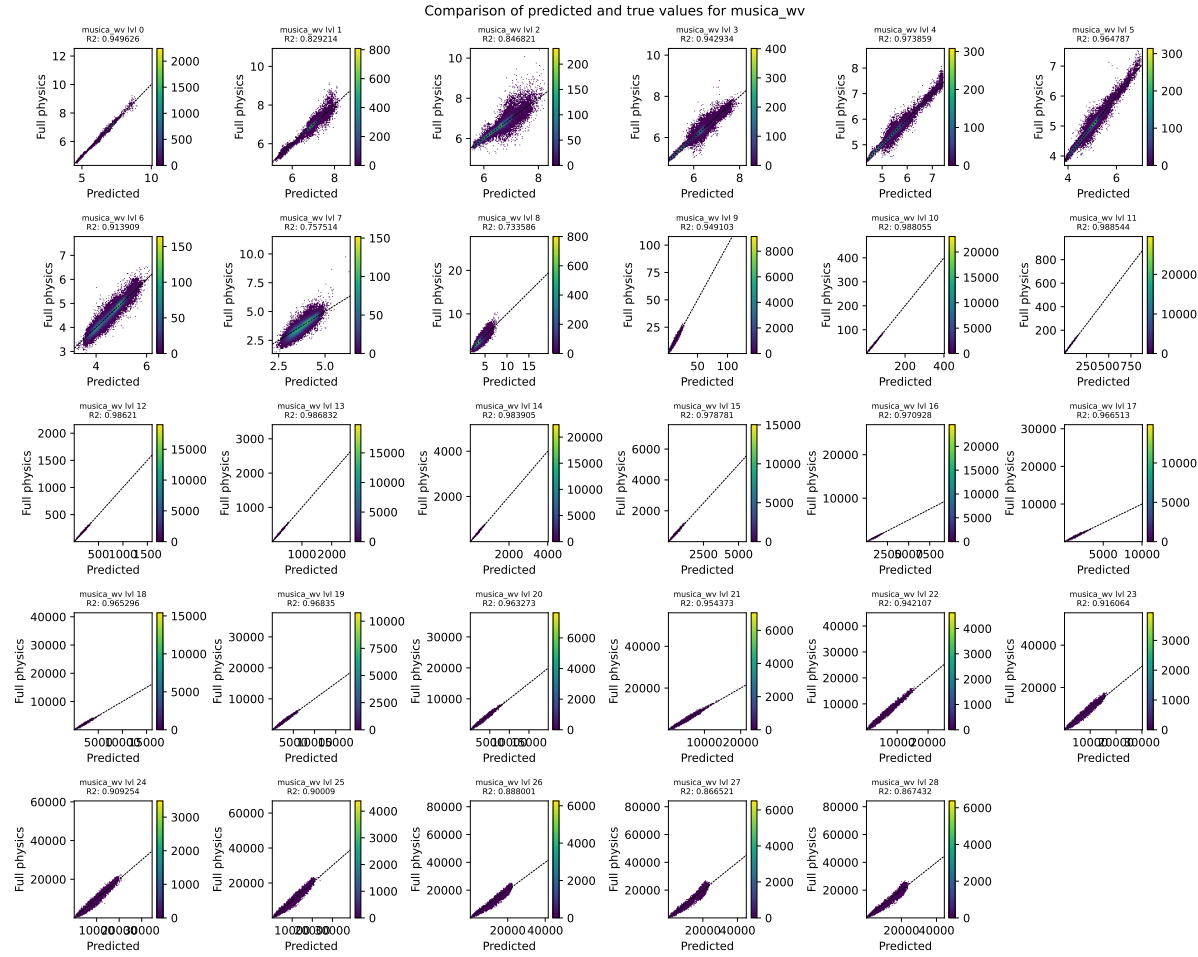


Figure 4.6.: Predicted over Actual Water Vapor Levels across atmospheric layers using Random Forest Regression. The average R^2 score of 0.925 927 is a great improvement over Ridge Regression. Again the levels 7 and 8 are the hardest to predict. Overall Random Forest Regression seems to struggle in similar atmospheric levels as Ridge Regression. The color scale shows points per pixel.

4.3. Gaussian Process Regression

In the exploration of Gaussian Process Regression (GPR) for atmospheric variable prediction, initial challenges were encountered and subsequent optimizations were made. Standard GPR attempts with only a Radial Basis Function (RBF) kernel were unsuccessful. However, improved results were achieved by employing a mixture of RBF, Matérn52, and white noise kernels.

The RBF kernel was used for smooth, infinitely differentiable functions, while the Matérn52 kernel allowed for less smooth functions. Measurement noise in the data was accounted for by the white noise kernel. Interestingly, minimal impact was observed when the RBF kernel was removed, with a slight decrease in performance for atmospheric temperature but improved predictions for atmospheric water vapor.

Model performance was enhanced through output normalization. The cubic space and time complexity of GPR initially limited the number of usable data points. To address this constraint, sparse GPR was investigated. Using 1000 randomly chosen inducing points allowed working with a larger dataset while maintaining computational feasibility. The random selection strategy seemed effective, due to a reasonably high number of inducing points. Alternatively, the locations of inducing points can be treated as hyperparameters and optimized along with other model parameters.

Lengthscales, determining input dimension relevance and function smoothness in each dimension, were found to play a crucial role in model performance. Initially, a single lengthscale was applied to all input dimensions. However, significant improvements were observed when individual lengthscales were assigned to each of the 950 input dimensions, enabling the model to capture varying importance and characteristics of different input features. Lengthscales were trained using normal GPR, as they did not change significantly from the starting lengthscales when learned with sparse GPR. This approach allowed for more accurate lengthscale estimation, which was then applied in the sparse GPR model.

Interestingly, doubling the total number of data points through the use of more powerful GPUs did not yield better results. The final GPR results were most impressive for water vapor. This is displayed in Figure 4.9. With an average R^2 value of 0.944 987, this represented a notable improvement over the Random Forest model (0.925 927), especially for levels seven and eight, as well as for all lower levels. However, challenges in accurately predicting water vapor in levels one and two persisted, suggesting room for further improvement in modeling the complex dynamics of water vapor in the upper atmosphere. Surface temperature prediction stayed excellent with an R^2 value of 0.999 720. For atmospheric temperature predictions, the model achieves an average R^2 value of 0.997 367, as shown in Figure 4.8. However, a slight decline in performance was observed, particularly when predicting extremely low temperatures in the upper troposphere. In these cases, the model tended to overestimate the atmospheric temperature, potentially indicating poor extrapolation to domains with few or no observations in the training data.

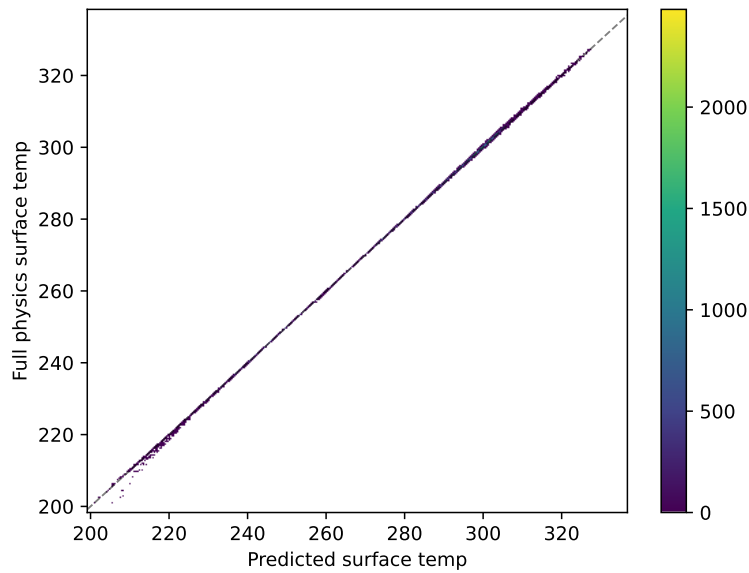


Figure 4.7.: Predicted over Actual Surface Temperature using GPR. The model achieves an R^2 score of 0.999 720, demonstrating excellent predictive performance for surface temperature. The color scale shows points per pixel.

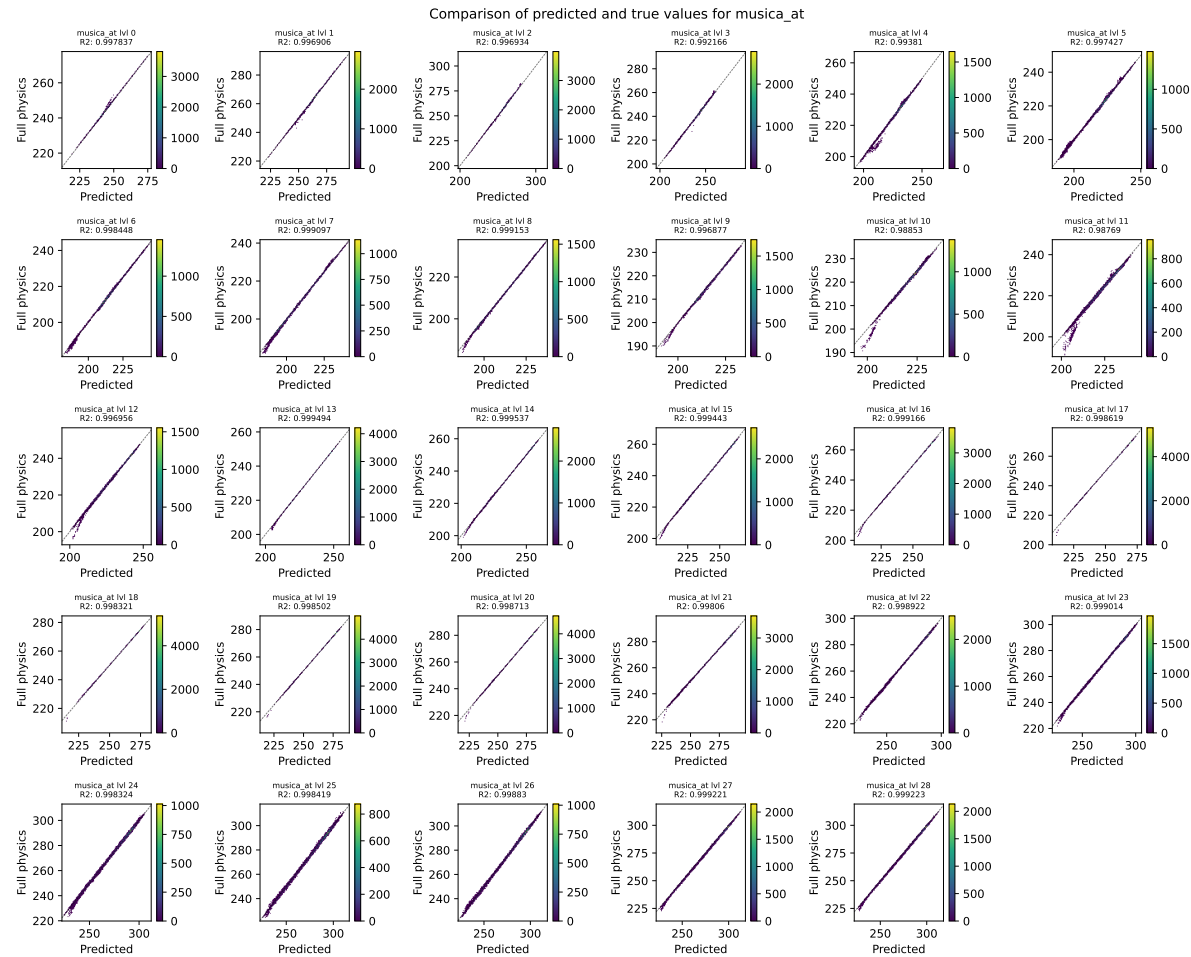


Figure 4.8.: Predicted over Actual Atmospheric Temperature across different atmospheric levels using GPR. The average R^2 score of 0.997 367 is a slight decline in performance. It can be seen when predicting extremely low temperatures, especially in the upper troposphere, where the model overestimates the atmospheric temperature. This could indicate poor extrapolation to domains with no or only few observations in the training data. The color scale shows points per pixel.

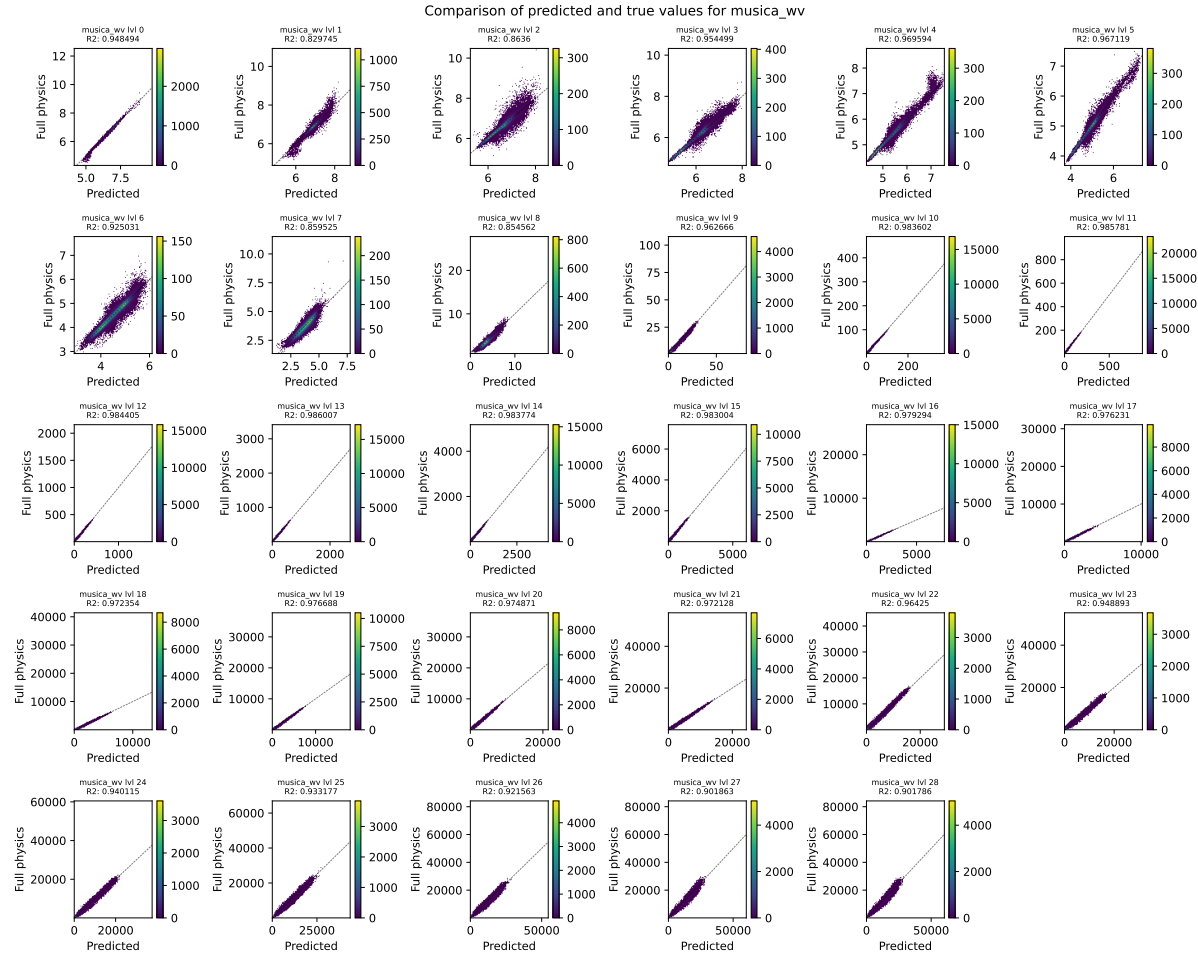


Figure 4.9.: Predicted over Actual Water Vapor Levels across atmospheric layers using GPR. The average R^2 score of 0.944 987 is a notable improvement over Random Forest. Especially the levels 7 and 8, as well as all lower levels, are being predicted a lot better. However, difficulties in levels 1 and 2 persist. The color scale shows points per pixel.

4.4. Graph Convolutional Networks

The initial motivation for exploring Graph Convolutional Networks (GCNs) stemmed from the hypothesis that additional information could be gained by considering the spatial and temporal context of observations. To test this hypothesis, a model comprising multiple GCNConv layers with $\text{ReLU}^{\text{Gloss}}$ as activation functions and a linear output layer was constructed.

Due to memory constraints, the graph was built in batches. Edges were drawn between observations based on their Euclidean distances in both space and time. If both distances fell below certain thresholds, a weighted edge was created, with the weight calculated as the sum of both distances.

Initial convergence challenges led to experimentation with various optimizers and schedulers, including Adam^{Gloss}, SGD with momentum 0 and 0.9, and Adagrad, in combination with step, plateau, one-cycle, and cosine annealing schedulers. The Adam optimizer with a cosine annealing scheduler was found to converge the fastest and most consistently. Further experimentation with network architecture revealed that three layers with 512 hidden states each produced the best results.

An investigation into different threshold factors for edge creation yielded an unexpected outcome. It was discovered that a factor of zero, effectively resulting in no edges in the graph, led to the best performance. This finding suggests that the observations do not benefit from additional spatial or temporal context. That is in line with the nature of the target variables derived from the full physics model that does not incorporate such contextual information. This as well as the influence of the number of layers can be seen in Figure 4.10.

With the removal of all edges, the GCN model demonstrated superior performance compared to GPR. After fine-tuning the model we obtained, average R^2 scores of 0.999 441 for surface temperature, 0.998 725 for atmospheric temperature, and 0.966 584 for water vapor profiles were achieved. These results indicate the model's strong predictive capabilities across all atmospheric variables. In Figure 4.13, we can see that there are still improvements to be made for water vapor predictions in the stratosphere. Otherwise Figures 4.11 and 4.12 show a tight clustering along the diagonal, performing well but not outperforming previous models.

In conclusion, while the initial hypothesis of leveraging spatial and temporal context through GCNs was not supported, the resulting model nevertheless proved highly effective in predicting atmospheric variables, especially improving water vapor predictions. This outcome underscores the complexity of atmospheric modeling and confirms the potential of neural network approaches.

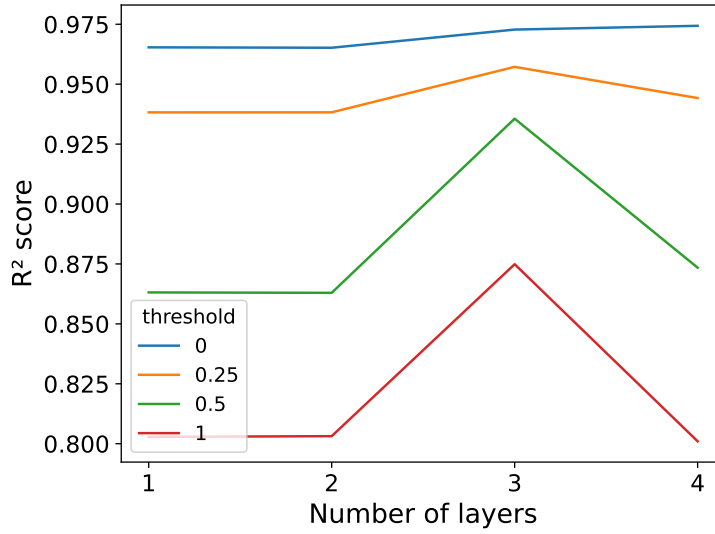


Figure 4.10.: Performance of GCN models with varying layer counts and edge configurations. For models with edges, performance peaks at three layers. The model without edges (isolated nodes) shows superior performance overall and benefits from additional layers, indicating that the network leverages node features effectively without relying on graph structure.

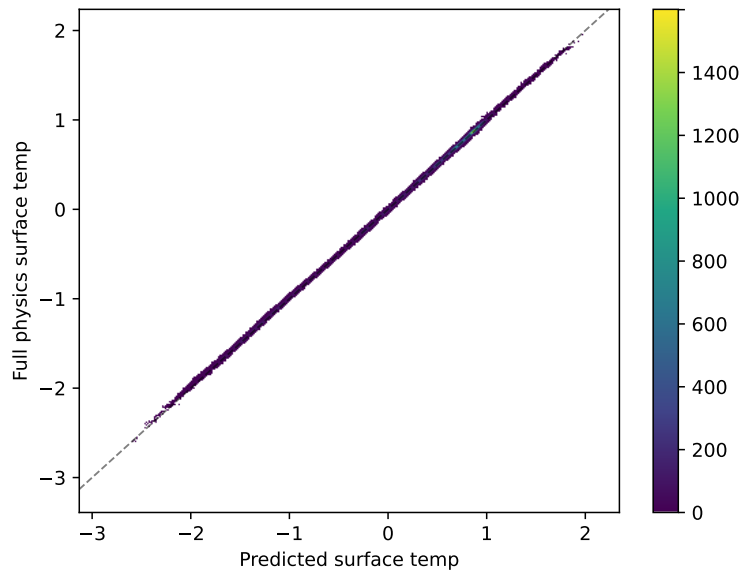


Figure 4.11.: Predicted over Actual Surface Temperature using GCN. The model achieves an R^2 score of 0.999 441, demonstrating excellent predictive performance for surface temperature. The color scale shows points per pixel.

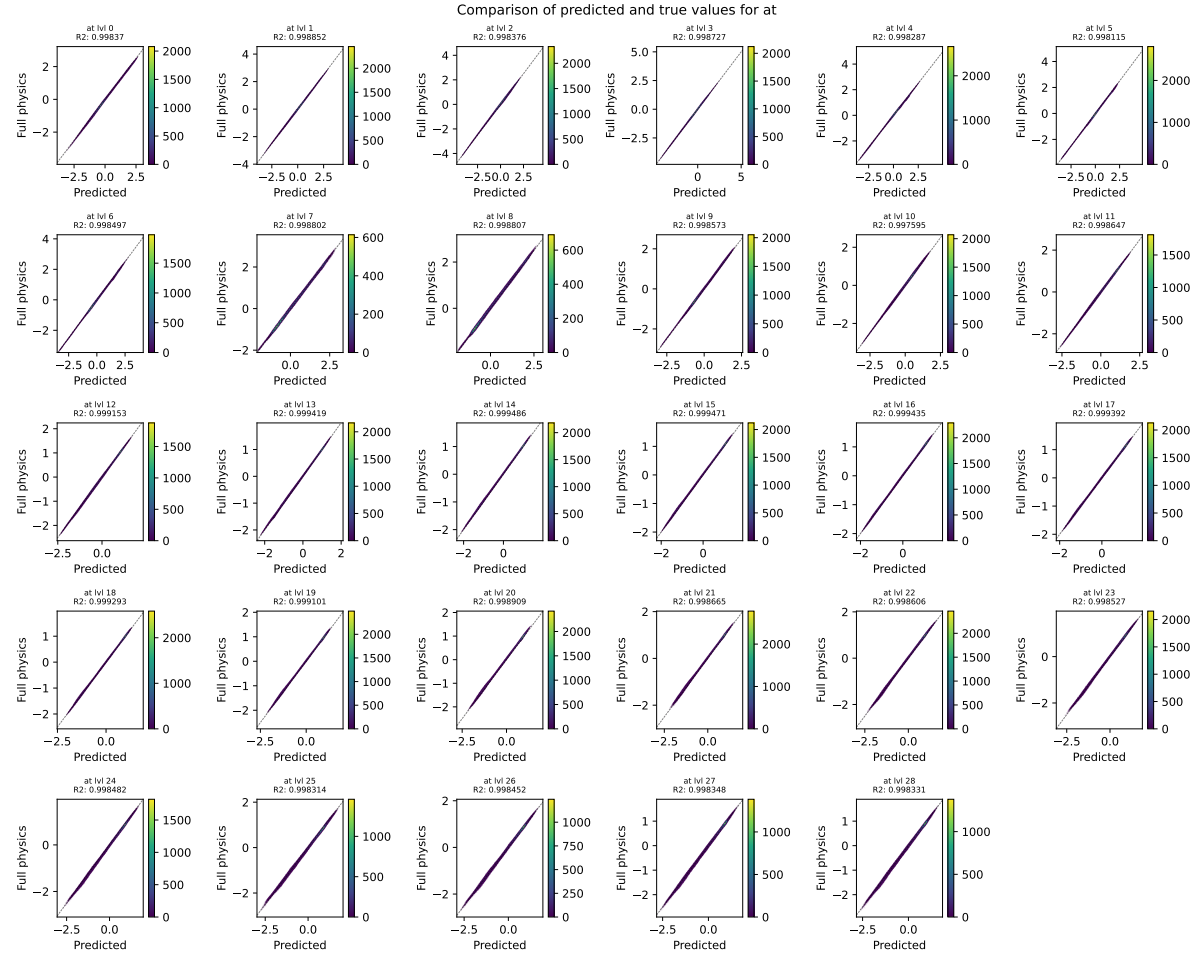


Figure 4.12.: Predicted over Actual Atmospheric Temperature across different atmospheric levels using GCN. The average R^2 score of 0.998 725 is still slightly worse than Ridge Regressions. No general biases can be seen, the overall performance is good, but worse than Random Forest. The color scale shows points per pixel.

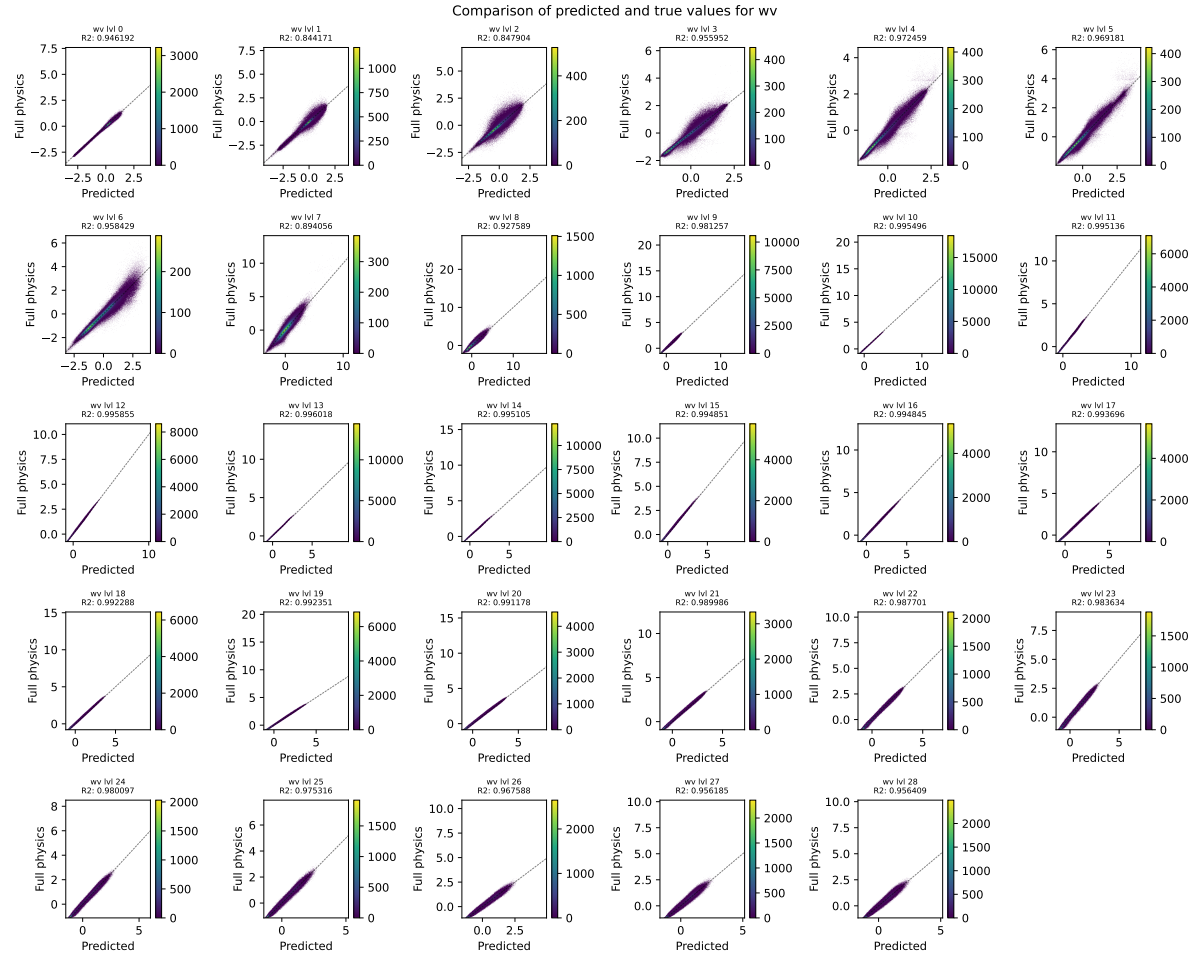


Figure 4.13.: Predicted over Actual Water Vapor Levels across atmospheric layers using GCN. The average R^2 score of 0.966 584 is a notable improvement over GPR. The levels most difficult to predict are still level 1 and 2, followed by level 7, continuing the trend of previous models. The color scale shows points per pixel.

4.5. Efficiency

To comprehensively compare the different models it is important to also examine the models' computational efficiency. The goal is to find a beneficial trade-off between the model's accuracy and computational efficiency. To compare the efficiency quantitatively three metrics have been used: total computation time, utilized CPU time i. e. sum of the time every core has been used, as well as energy consumed by the node when making predictions for observations from one orbit. To make the results as comparable as possible all evaluations happened on a CPU only node on HoreKa (2 Intel Xeon Platinum 8368, 76 cores, 152 threads, 256 GB of memory). The output of each job is shown in Table 4.1. The results show clearly that all the machine learning methods are significantly more efficient than the physical processor.

The machine learning methods are more efficient by three to five orders of magnitude in terms of energy consumption and two to four orders of magnitude faster in terms of total job wall time. These impressive results suggest that processing could likely even be done on a lightweight machine. Expectedly, Ridge Regression is most efficient, followed by the overall best performing model GCN, as well as GPR and Random Forest. Interestingly, the CPU time for GPR is higher than its total processing time. This discrepancy is due to the parallelized implementation, which utilizes multiple cores simultaneously. The same applies to the energy consumption, the power draw of a node running at all is higher than the incremental consumption of using more cores on the same node. It should also be noted that about 50 seconds of the job runtime is constant and only serves the preparation of the node i. e. no user code is executed during this time, so predicting more orbits should result in lower times per orbit.

However, it is crucial to note that the numbers for the physical processor are not entirely representative. The full physics processor calculates additional outputs, such as trace gases, which contributes to its longer processing time, but this does not account for the majority of the difference in efficiency. The fundamental reason for the physical processor's higher computational cost lies in the radiative transfer^{Gloss.} calculations required to understand the measured spectra. This step is computationally expensive and necessary even when processing only water vapor and atmospheric temperature.

When considering the overall efficiency of these methods, training times for the models should not be completely neglected. Ridge Regression training is typically very fast, while Random Forest, GPR and GCN training can be quite slow. Approximations for the training times of the models, without hyperparameter search, are shown in Table 4.2. However, they should be interpreted with great caution, as they are highly dependent on model size, dataset size, implementation of the model, and whether GPU implementations are available and used. Additionally, the process of finding the right hyperparameters can be computationally expensive, which further complicates the efficiency comparison.

Table 4.1.: Table showing different measured metrics for the different models, when evaluating a single orbit. It is clear that the machine learning methods are much more efficient.

	Physical Processor	Ridge Regression	Random Forest	GPR	GCN
Job wall time [hh:mm:ss]	07:02:03	00:05 (00:50)*	00:49 (08:05)*	00:31 (05:10)*	00:06 (01:03)*
CPU Utilized [dd-hh:mm:ss]	31-16:22:00	00:00 (00:03)*	00:43 (07:10)*	05:09 (51:34)*	00:01 (00:11)*
Energy used [Wh]	4122.80	0.05 (0.51)*	3.66 (36.61)*	2.43 (24.31)*	0.12 (1.22)*

* orbit was predicted 10 times; average (total)

Table 4.2.: Table showing estimated values for the training of different models. These numbers are based on a single training run, without searching for hyperparameters and without fine-tuning. These numbers are not very well suited for comparison, due to very different frameworks, being used for the different models, the size of the training set differing, and some of the models being trained on GPU.

	Ridge Regression	Random Forest	GPR	GCN
Job wall time [hh:mm:ss]	0:10:00	8:00:00	0:10:00*	1:00:00*
CPU Utilized [d-hh:mm:ss]	1-00:00:00	52-00:00:00	0:02:00*	00:55:35*
Energy used [Wh]	50	5000	100*	600*

* training utilized GPUs making the numbers not comparable

4.6. Resource Management

Efficient resource management was crucial for optimizing the performance on the High Performance Computing (HPC) system HoreKa. Given the scale of the datasets and the complexity of the models, careful consideration was required in selecting the appropriate resources to balance computational efficiency and memory utilization.

4.6.1. Data Management and Node Selection

A key aspect of the resource management was determining the amount of data to use for each experiment, in order to keep computing times reasonable and use (the limited) large memory nodes only when really necessary, as well as making sure not to run Out Of Memory (OOM) on the nodes themselves or on the GPU(s).

4.6.2. Utilization of CPU Cores

Given that the HPC nodes were allocated on an exclusive basis, it was essential to utilize as many CPU cores as possible to accelerate computation. Tasks were parallelized wherever possible e.g. by training multiple models in parallel when optimizing hyperparameters. This reduced the overall runtime, making additional experiments possible.

4.6.3. Maximizing GPU Utilization

The effective use of GPU resources was another critical factor in reducing computation time. As GPUs are a limited resource on HoreKa it was made sure to only use them when they had a significant advantage over CPUs. Not all models were suited for multiple GPUs, but when possible all GPUs were utilized, sometimes by training multiple models in parallel.

4.6.4. Memory Mapping and Efficient Data Loading

Memory efficiency was a major consideration, especially when handling large datasets that could not be fully loaded into memory. For tasks that involved preprocessing or operations requiring only a subset of the data, memory mapping techniques were employed. Mapping data directly from disk to memory allowed us to load data into memory only when it was needed, significantly reducing the memory footprint. Additionally, the storage format of the datasets was optimized to minimize the data loading times. This was done by saving the data merged in binary formats. This reduced the overhead associated with data I/O, thereby speeding up the overall processes.

5. Conclusion

The results demonstrate significant efficiency gains compared to the optimal estimation by the full physics processor, being three to five orders of magnitude more efficient in terms of energy consumption and two to four orders of magnitude faster in terms of total job wall time. These results indicate that machine learning methods can potentially partially replace full physics processing in atmospheric data analysis.

The machine learning models developed in this work offer significantly faster data processing, potentially enabling real-time analysis and reducing the required computational resources for atmospheric modeling while maintaining high accuracy in predicting key atmospheric variables. When predicting temperature variables, the simplest model Ridge Regression did best, while the more complex models performed better on water vapor predictions, with Graph Convolutional Network (GCN) performing best.

The improved efficiency and speed of these machine learning methods open up new possibilities in various applications, notably in numerical weather prediction. Real-time processing of atmospheric data could be achieved, making this data available for weather forecasting. Additionally, once the models are trained, processing the data does not require server-grade hardware anymore and could likely even be done on a lightweight machine.

These advantages come at the cost of a reduction in performance. Moreover, machine learning methods present greater challenges in interpretation compared to physical models. Neural Networks, in particular, require post-hoc methods for explanation. The lack of explainability in these models leads to unknown uncertainties and uncertainty regarding their extrapolation capabilities.

While full physics models remain crucial for comprehensive atmospheric understanding, the integration of machine learning techniques, as demonstrated in this thesis, represents a significant step towards more efficient and responsive atmospheric data processing systems.

5.1. Future Work

Several avenues for future research have been identified to build upon the findings of this thesis. The scope could be extended to include additional retrieval target variables from the Infrared Atmospheric Sounding Interferometer (IASI)^{Gloss.} dataset, namely greenhouse gases such as N_2O , CH_4 , and HNO_3 .

Given the demonstrated accuracy of GCNs, other neural network approaches could be explored. Interestingly, no accuracy improvements came from additional spatio-temporal context. Approaches without a graph structure might lead to more efficient processing and potentially more accurate predictions.

To leverage long-term or global relationships within atmospheric data, the investigation of transformer architectures could be a promising direction. Transformers have shown remarkable capabilities in modeling complex, long-range dependencies in various domains, and their application to atmospheric data analysis could yield valuable insights.

These future research directions aim to further enhance the accuracy and efficiency of machine learning in atmospheric measurements, potentially leading to significant advancements in our ability to understand and predict complex atmospheric phenomena.

Acknowledgement

This research was made possible through the generous provision of MUSICA IASI^{Gloss} full retrieval product [20] by the Institute of Meteorology and Climate Research - Atmospheric Trace Gases and Remote Sensing (IMK-ASF) at the Karlsruhe Institute of Technology (KIT).

This work was performed on the HoreKa supercomputer funded by the Ministry of Science, Research and the Arts Baden-Württemberg and by the Federal Ministry of Education and Research.

The implementation of various machine learning models in this work relied on several open-source software packages. We acknowledge the use of the following libraries:

Scikit-learn [15] for implementing Ridge Regression and Random Forest models

GPflow [13] for Gaussian Process Regression

PyTorch Geometric [5] for Graph Convolutional Network models

These tools were essential in the development and evaluation of the machine learning approaches presented in this thesis. Their robustness and efficiency significantly contributed to the successful completion of this research.

Acronyms

AMSU-A Advanced Microwave Sounding Unit - A 5

CART Classification and Regression Trees 16

EUMETSAT European Organisation for the Exploitation of Meteorological Satellites 2, 5, 45

GCN Graph Convolutional Network i, ii, 3, 15, 17–19, 31–36, 39, 40

GNN Graph Neural Network 17

GPR Gaussian Process Regression i, ii, 3, 15–17, 19, 27–31, 34–36, 51

HPC High Performance Computing 36, 37

IASI Infrared Atmospheric Sounding Interferometer i, ii, 1, 5, 19, 39, 41, *Glossary*: IASI

Metop Meteorological Operational 5

MHS Microwave Humidity Sounder 5

MUSICA Multi-platform remote Sensing of Isotopologues for investigating the Cycle of Atmospheric water 5, 10, 14, 41

OLS Ordinary Least Squares 15

OOM Out Of Memory 36

ppm parts per million 11, 12

RBF Radial Basis Function 17, 27

SST Sea Surface Temperatures 2

Glossary

Adam An algorithm for first-order gradient-based optimization based on adaptive estimates of lower-order moments [10]. 31

Hyperparameter Parameters which specify details of the learning process and can not be learned. 15–17, 23, 27, 35–37

Infrared Atmospheric Sounding Interferometer (IASI) IASI is the core instrument of the currently three Metop satellites of the European Organization for the Exploitation of Meteorological Satellites European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) [21]. i, ii, 1–3, 5, 10, 19, 39, 41

Polar Orbit A polar orbit is an orbit where the satellite almost passes both poles in contrast to a near-equatorial orbit. 7

Radiative Transfer Radiative transfer describes the propagation of radiation through matter, it can be used to determine chemical composition and temperature in the atmosphere. 2, 35

Rectified Linear Unit ReLU keeps all positive values and sets negative values to zero

$$\text{ReLU}(x) = \max(0, x) .$$

18, 31

Singular Value Decomposition Decomposition of a matrix $X \in \mathbb{R}^{m \times n}$ into $X = U\Sigma V^*$ where U and V are orthogonal, V^* is the complex conjugate of V and Σ is a diagonal matrix 15

Zenith Here, the vector from the ground to the sky directly above. 6, 8

Bibliography

- [1] *About IASI*. en. URL: https://www.esa.int/Applications/Observing_the_Earth/Meteorological_missions/MetOp/About_IASI (visited on 08/27/2024).
- [2] Alessandro Cotronei and Thomas Slawig. “Single-precision arithmetic in ECHAM radiation reduces runtime and energy consumption”. en. In: *Geoscientific Model Development* Volume 13 (June 2020), pp. 2783–2804. ISSN: 1991-9603. DOI: 10.5194/gmd-13-2783-2020.
- [3] Mark Ebden. *Gaussian Processes: A Quick Introduction*. en. Aug. 2015. URL: <http://arxiv.org/abs/1505.02965> (visited on 09/04/2024).
- [4] *EUMETSAT - Product Navigator - IASI Level 2 - Temperature and Humidity Profiles - Metop - Regional Data Service*. URL: <https://navigator.eumetsat.int/product/EO:EUM:DAT:METOP:EARS-IASI-L2> (visited on 08/25/2024).
- [5] Matthias Fey and Jan Eric Lenssen. *Fast Graph Representation Learning with PyTorch Geometric*. May 2019. URL: https://github.com/pyg-team/pytorch_geometric (visited on 09/13/2024).
- [6] Bruno Franco, Lieven Clarisse, Trisevgeni Stavrakou, Jean-Francois Müller, Martin Van Damme, Simon Whitburn, Juliette Hadji-Lazaro, Daniel Hurtmans, Domenico Taraborrelli, Cathy Clerbaux, and Pierre Francois Coheur. “A General Framework for Global Retrievals of Trace Gases From IASI: Application to Methanol, Formic Acid, and PAN”. en. In: *Journal of Geophysical Research: Atmospheres* Volume 123 (2018), pp. 13, 963–13, 984. ISSN: 2169-8996. DOI: 10.1029/2018JD029633.
- [7] *Gaussian Processes*. URL: <https://cs.stanford.edu/~rpryzant/blog/gp/gp.html> (visited on 09/04/2024).
- [8] Pierre Geurts, Damien Ernst, and Louis Wehenkel. “Extremely Randomized Trees”. In: *Machine Learning* Volume 63 (Apr. 2006), pp. 3–42. DOI: 10.1007/s10994-006-6226-1.
- [9] Trevor Hastie. “Ridge Regularization: An Essential Concept in Data Science”. In: *Technometrics : a journal of statistics for the physical, chemical, and engineering sciences* Volume 62 (2020), pp. 426–433. ISSN: 0040-1706. DOI: 10.1080/00401706.2020.1791959.
- [10] Diederik P. Kingma and Jimmy Ba. *Adam: A Method for Stochastic Optimization*. Jan. 2017. DOI: 10.48550/arXiv.1412.6980. URL: <http://arxiv.org/abs/1412.6980> (visited on 09/10/2024).
- [11] Thomas N. Kipf and Max Welling. *Semi-Supervised Classification with Graph Convolutional Networks*. en. Feb. 2017. URL: <http://arxiv.org/abs/1609.02907> (visited on 08/21/2024).
- [12] Otto M. Lamminpää, Jouni I. Susiluoto, Jonathan M. Hobbs, James L. McDuffie, Amy J. Braverman, and Houman Owahdi. “Forward Model Emulator for Atmospheric Radiative Transfer Using Gaussian Processes And Cross Validation”. English. In: *Atmospheric Measurement Techniques Discussions* (May 2024), pp. 1–32. DOI: 10.5194/amt-2024-63.

- [13] Alexander G. de G. Matthews, Mark van der Wilk, Tom Nickson, Keisuke Fujii, Alexis Boukouvalas, Pablo Le{\'o}n-Villagr{\'a}, Zoubin Ghahramani, and James Hensman. “GPflow: A Gaussian Process Library using TensorFlow”. In: *Journal of Machine Learning Research* Volume 18 (2017), pp. 1–6. ISSN: 1533-7928.
- [14] *Optuna - A hyperparameter optimization framework*. en. URL: <https://optuna.org/> (visited on 09/07/2024).
- [15] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. “Scikit-learn: Machine Learning in Python”. In: *Journal of Machine Learning Research* Volume 12 (2011), pp. 2825–2830. ISSN: 1533-7928.
- [16] Joaquin Quiñonero-Candela and Carl Edward Rasmussen. “A Unifying View of Sparse Approximate Gaussian Process Regression”. In: *Journal of Machine Learning Research* Volume 6 (2005), pp. 1939–1959. ISSN: 1533-7928.
- [17] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian processes for machine learning*. en. Cambridge, Mass: MIT Press, 2006. ISBN: 978-0-262-18253-9.
- [18] Tao Ren, Michael F. Modest, Alexander Fateev, Gavin Sutton, Weijie Zhao, and Florin Rusu. “Machine learning applied to retrieval of temperature and concentration distributions from infrared emission measurements”. In: *Applied Energy* Volume 252 (Oct. 2019), p. 113448. ISSN: 0306-2619. DOI: [10.1016/j.apenergy.2019.113448](https://doi.org/10.1016/j.apenergy.2019.113448).
- [19] Stéphane Saux Picart, Pierre Tandeo, Emmanuelle Autret, and Blandine Gausset. “Exploring Machine Learning to Correct Satellite-Derived Sea Surface Temperatures”. en. In: *Remote Sensing* Volume 10 (Feb. 2018), p. 224. ISSN: 2072-4292. DOI: [10.3390/rs10020224](https://doi.org/10.3390/rs10020224).
- [20] Matthias Schneider. *KIT-IMK-ASF - About IMK-ASF - Research Groups - Ground-Based Remote Sensing - MUSICA - Data - MUSICA IASI v030201*. en-gb. Feb. 2023. URL: <https://www.imk-asf.kit.edu/english/musica-iasi-v321.php> (visited on 05/15/2024).
- [21] Matthias Schneider, Benjamin Ertl, Christopher J. Diekmann, Farahnaz Khosrawi, Andreas Weber, Frank Hase, Michael Höpfner, Omaira E. García, Eliezer Sepúlveda, and Douglas Kinnison. “Design and description of the MUSICA IASI full retrieval product”. English. In: *Earth System Science Data* Volume 14 (Feb. 2022), pp. 709–742. ISSN: 1866-3508. DOI: [10.5194/essd-14-709-2022](https://doi.org/10.5194/essd-14-709-2022).
- [22] Matthias Schneider and Frank Hase. “Optimal estimation of tropospheric H₂O and δD with IASI/METOP”. English. In: *Atmospheric Chemistry and Physics* Volume 11 (Nov. 2011), pp. 11207–11220. ISSN: 1680-7316. DOI: [10.5194/acp-11-11207-2011](https://doi.org/10.5194/acp-11-11207-2011).
- [23] Peter Ukkonen. “Exploring Pathways to More Accurate Machine Learning Emulation of Atmospheric Radiative Transfer”. en. In: *Journal of Advances in Modeling Earth Systems* Volume 14 (2022), e2021MS002875. ISSN: 1942-2466. DOI: [10.1029/2021MS002875](https://doi.org/10.1029/2021MS002875).

A. Appendix

The code used to produce the results in this thesis is available in a GitHub repository at <https://github.com/davidvigneron/Processing-State-of-the-Art-Satellite-Measurements-with-Machine-Learning> or on the GitLab of the Institute at https://gitlab.kit.edu/peer.nowack/kit-ai-climate-group-projects/-/tree/master/bachelor_projects/david_vigneron/Processing-State-of-the-Art-Satellite-Measurements-with-Machine-Learning. This repository includes scripts used for data preprocessing, model implementation, and result analysis.

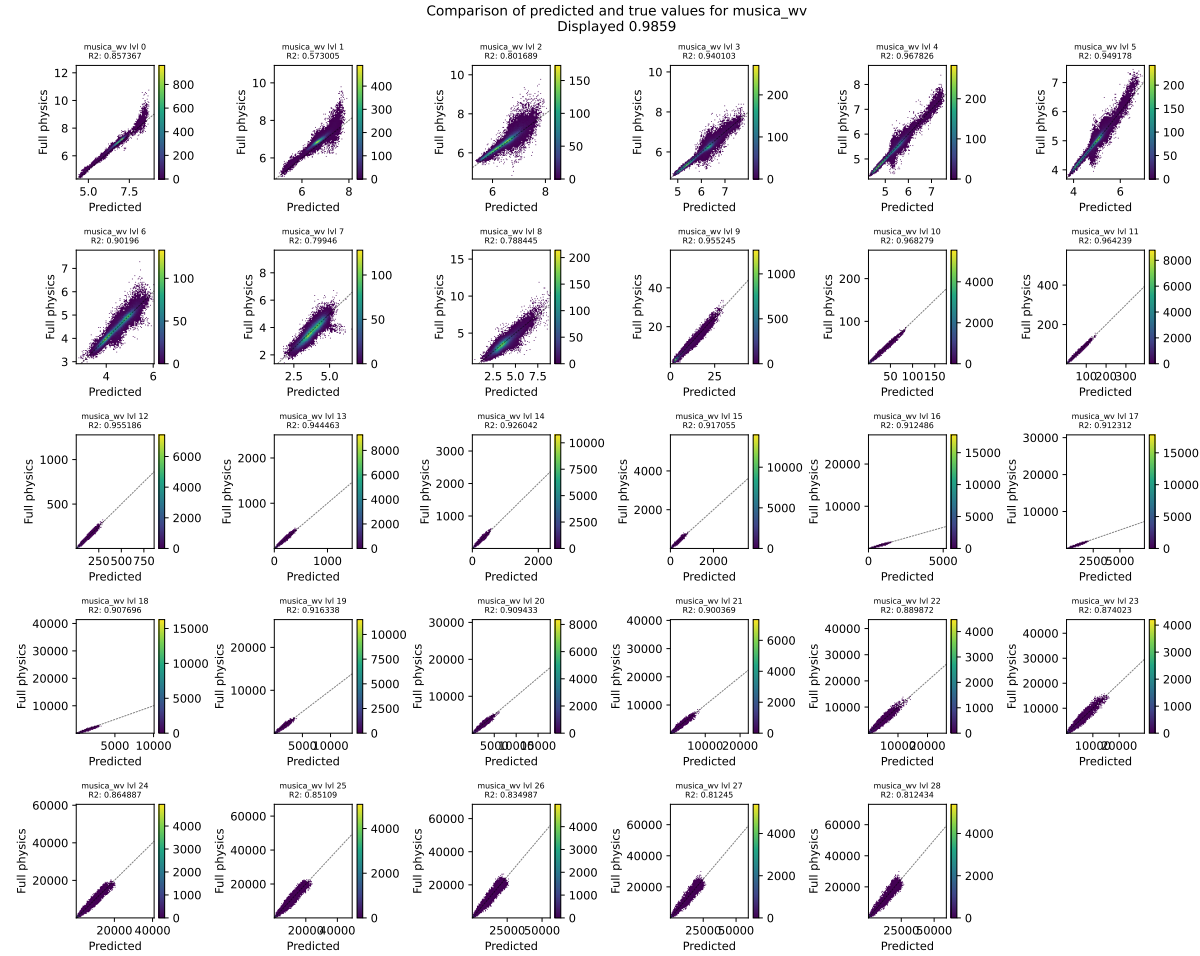


Figure A.1.: Comparison of full physics processor output to predicted values for Ridge Regression when applying a log transformation to all water vapor values (incl. a priori). This model performed better at the atmospheric levels 7 to 17. The log transformation seemed to cause some rare but extreme outliers, which is why only 98.59% of the points are shown. This is annoying and means there is need for edge case specifications when applying such transformation. This issue highlights the necessity for edge case specifications when applying such transformations.

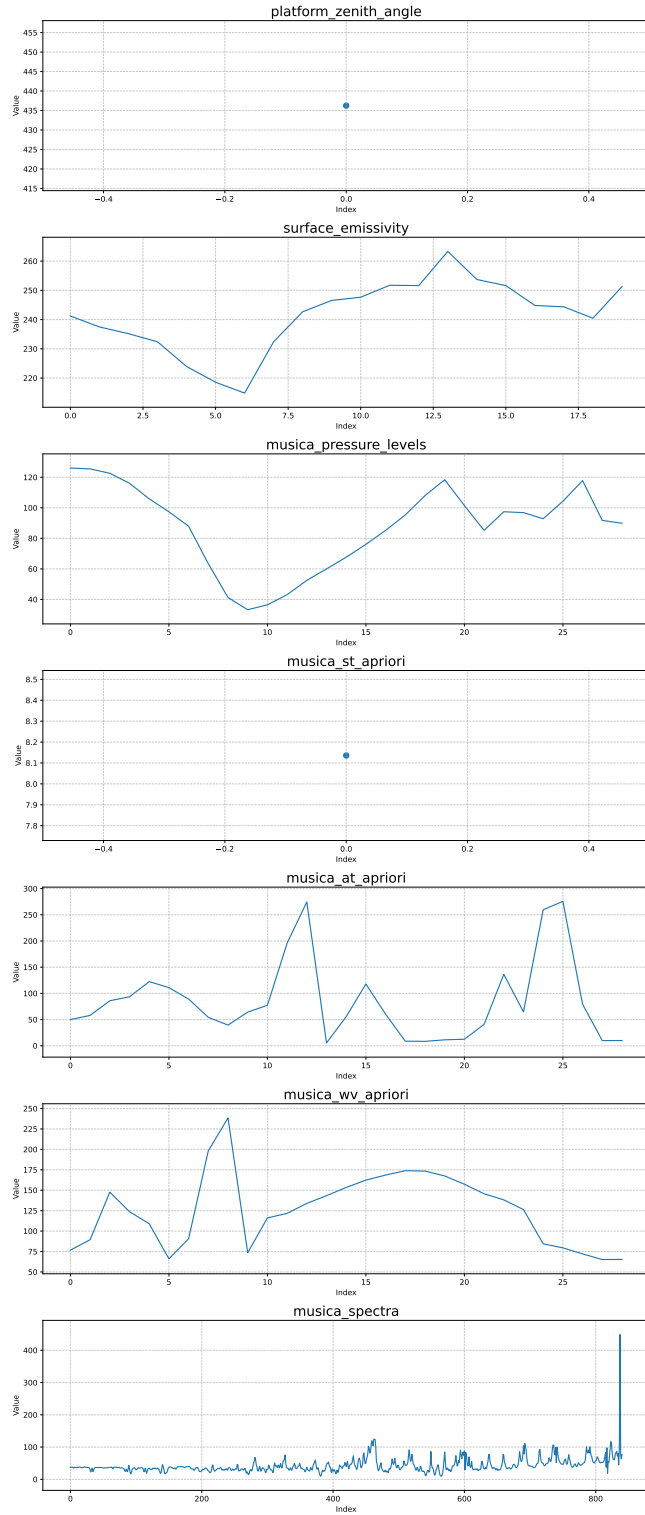


Figure A.2.: Final lengthscales for all input variables in Gaussian Process Regression (GPR)